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Leveraging large language models to enhance multi-agent risk assessment in supply chain networks

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ABSTRACT

We propose a novel large language model (LLM) enhanced framework, MARS (Multi-Agent Risk assessment in Supply chain networks), for risk assessment and integration of both structured and unstructured factors for logistic hub site selection across a target territory, using the southeastern U.S. states as a testbed. While structured factors such as cost, distance, traffic accidents, traffic congestion, and crime rates can be directly computed, unstructured severe weather event narratives need to be interpreted semantically through LLMs. We introduce a multi-agent architecture featuring three specialised agents, *RiskAgent*, *FeedbackAgent*, and *RevisionAgent*, that collaborate through a feedback-revision loop to convert raw extreme weather event narratives into fine-grained risk severity levels. By integrating these severity levels with structured indicators via aggregation, the proposed method enables interpretable and risk-aware ranking of candidate hubs, thereby supporting informed decision-making for logistic hub site selection.

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Generative AI; large language models; multi-agent system; supply chain; risk assessment; decision-making

1. Introduction

Modern supply chains operate under increasingly volatile, uncertain, complex, and ambiguous (VUCA) conditions. As global supply networks become more interconnected, the structural interdependencies among suppliers, logistics providers, infrastructure nodes, and digital systems intensify the exposure to systemic vulnerabilities. In such tightly coupled systems, disruptions are no longer isolated; instead, they propagate across organisational and geographic boundaries through physical, informational, and financial linkages (Ivanov 2022; Katsaliaki, Galetsi, and Kumar 2022). Local disturbances, arising from natural hazards, cyber vulnerabilities, regulatory shifts, or socio-political instability, can rapidly escalate into large-scale operational failures and strategic bottlenecks.

Interconnected supply chain networks (Kulkarni, Dahan, and Montreuil 2022; Montreuil 2011; Muthukrishnan, Kulkarni, and Montreuil 2024) face a broad spectrum of risks originating from both internal operations and external environments. Among others, these risks include disruptions in supply and demand, financial volatility, cyber attacks, regulatory changes, and the rapid spread of misinformation (Chu, Park, and Kremer 2020).

They rarely occur in isolation; instead, they often interact and compound one another, undermining the stability of end-to-end supply chains. More specifically, demand surges or supply shortages can lead to production delays and stockouts; financial shocks may reduce supplier reliability or trigger contract breaches; cyber incidents can paralyze logistics systems and compromise data integrity; and policy changes or trade restrictions can disrupt cross-border flows. Meanwhile, inaccurate or delayed information can impair situational awareness and lead to suboptimal responses. When coordination is weak or system buffers are limited, even small disturbances can escalate into significant disruptions, affecting inventory availability, lead time reliability, and customer service levels. These cascading effects are especially critical in location-dependent decisions such as logistic hub placement. Regional exposure to compound risks such as weather disruptions, infrastructure fragility, or crime can directly affect long-term operational resilience and service continuity.

Despite the growing complexity of supply chain risks, many traditional risk assessment frameworks remain reliant on structured data, which we use to refer to numerical or categorical indicators such as crime,

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bankruptcy, strike, or traffic. These approaches often fail to capture the contextual richness, interdependencies, and fast-evolving nature of disruptions, especially when risks stem from unstructured sources such as textual event reports, news narratives, or social media feeds (Emrouznejad, Abbasi, and Sıcakyüz 2023; Fosso Wamba et al. 2024). In high-stakes scenarios such as logistic hub placement, where decision-makers must balance exposure to regional risks with operational costs and service coverage, there is a growing need for more adaptive, intelligent, and interpretable decision-making frameworks. Recent advancements in artificial intelligence (AI), particularly in large language models (LLMs) and multi-agent systems (MAS), offer promising opportunities to bridge this gap by enabling the integration of heterogeneous data sources and supporting more contextual, proactive risk analysis.

While LLMs and MAS have recently shown potential in supply chain applications, their use in fine-grained, risk-aware planning remains limited. In particular, high-stakes tasks such as logistic hub site selection, which require synthesising structured data and dynamic, context-rich risk indicators, have received little attention. Prior work (Giannakis and Louis 2011; Ivanov, Sokolov, and Dolgui 2014; Klibi, Martel, and Guitouni 2010) often addresses general decision support or focuses narrowly on structured metrics, lacking integrated frameworks capable of processing unstructured narratives alongside traditional indicators in a transparent and adaptive manner.

In this paper, we propose a novel LLM-enhanced multi-agent framework, MARS (Multi-Agent Risk assessment in Supply chain networks), for risk-informed logistic hub site selection. MARS combines structured risk indicators (e.g. cost, distance, traffic accidents, traffic congestion, and crime rates) with unstructured severe weather event narratives, using real-world location and property datasets covering the southeastern United States in our testbed. The framework introduces three specialised agents, namely *RiskAgent*, *FeedbackAgent*, and *RevisionAgent*, which collaborate through a feedback-revision loop to convert narrative reports into quantitative severity assessments. While structured data are directly computed and normalised, unstructured narratives are semantically interpreted using LLMs to support fine-grained risk analysis. Candidate hubs are then ranked using a composite scoring mechanism that aggregates all risk indicators. MARS enables systematic, interpretable, adaptive, and iterative assessment of diverse risk factors, providing robust decision support for selecting resilient logistic hub locations. An overview of the MARS framework is illustrated in Figure 1.

In this study, we aim to address the following research questions (RQs):

- **RQ1:** How to design a LLM-based framework for supply chain risk assessment?
- **RQ2:** How to coordinate multiple LLM agents to interpret unstructured risks?
- **RQ3:** How to aggregate both structured and unstructured risk factors?

Answering these research questions, our main contributions in this paper are as follows:

- (1) **LLM-based framework for supply chain risk assessment:** We propose a novel framework, MARS, that leverages LLMs to enable adaptive, context-aware, and proactive risk assessment, supporting resilient decision-making in supply chain networks. Specifically, we apply MARS to logistic hub site selection across southeastern U.S. states as a testbed, demonstrating its capacity to support resilient infrastructure planning under risk conditions.
- (2) **Multi-agent coordination to interpret unstructured risks:** We design a multi-agent system that distributes tasks such as extraction, reasoning, validation, and refinement across specialised agents, enabling collaborative processing of unstructured risk information. Through dynamic interaction and feedback-revision loop mechanisms, the system progressively improves its outputs, yielding more reliable and robust risk evaluations.
- (3) **Aggregation of LLM-driven unstructured risk analysis with structured risk factors:** We integrate insights extracted from unstructured data by LLM agents with structured indicators, enabling a more comprehensive and data-rich basis for logistic hub evaluation. This aggregation provides a unified foundation for balancing diverse risk considerations in decision-making.

2. Related work

In this section, we review prior work relevant to our study. We begin with an overview of supply chain risk management (SCRM), including major research directions and known limitations. We then discuss recent applications of large language models (LLMs) in the supply chain domain, emphasising their roles in decision support and knowledge extraction. Finally, we review the emerging literature on LLM-based multi-agent systems (MAS), outlining how these approaches can improve collaboration, reasoning, and adaptability in SCRM contexts.

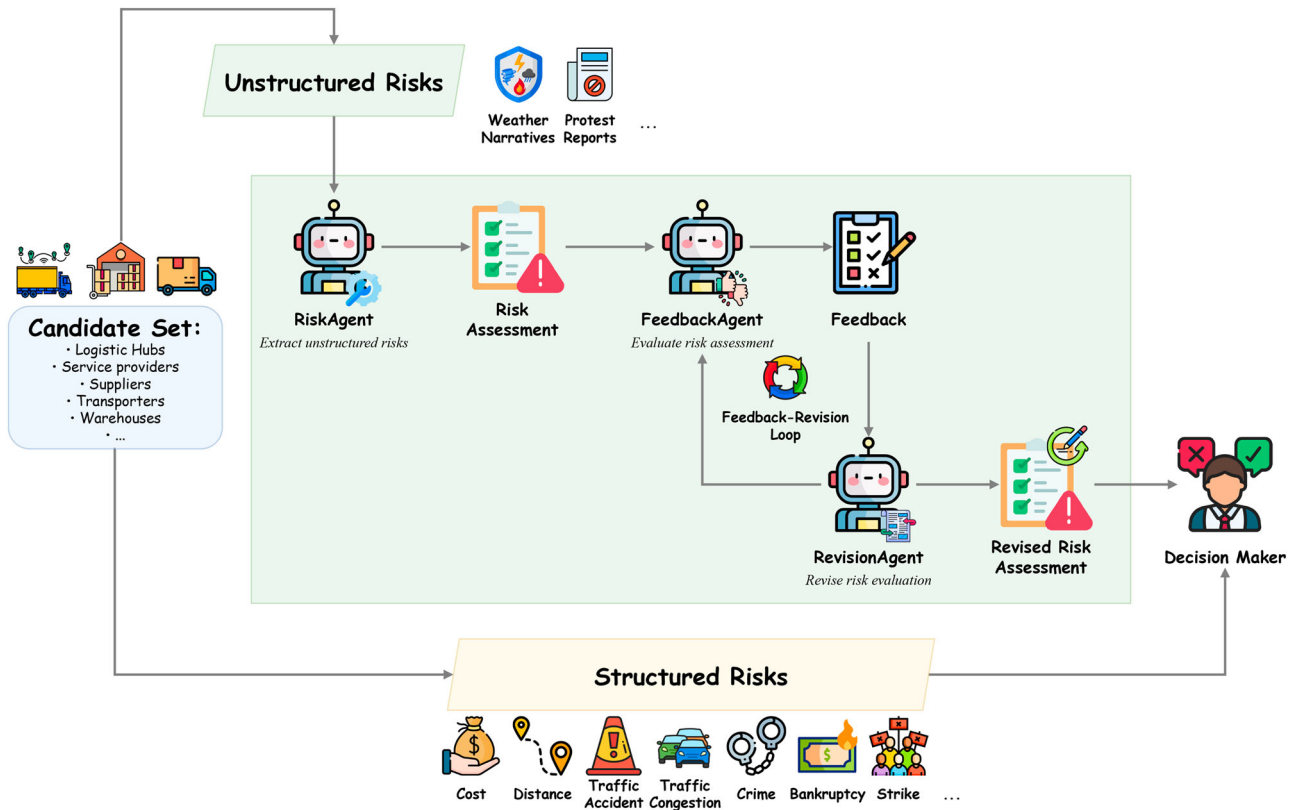


Figure 1. MARS framework for LLM-enhanced multi-agent risk assessment in supply chain networks.

2.1. Supply chain risk management

The operational landscape of contemporary supply chains is increasingly shaped by volatility, uncertainty, complexity, and ambiguity (VUCA) (Wilding 1998). Both sudden disruptions and systemic shocks threaten network stability and complicate strategic decision-making. These factors make SCRM a critical focus, emphasising early detection of disruptions, systematic evaluation of vulnerabilities, and the design of strategies to safeguard global supply networks (Ponomarov and Holcomb 2009). Yet implementing SCRM in VUCA environments is difficult. The high level of interconnectivity makes supply chains particularly vulnerable to disruptions from natural disasters, geopolitical tensions, cyber attacks, and economic shocks. Recent studies further report a growing frequency and severity of such disruptions, which can propagate across networks and amplify operational and financial risks (Ivanov 2022; Katsaliaki, Galetsi, and Kumar 2022). These trends reveal the limitations of traditional risk management approaches that often rely on static models and lack real-time adaptability.

SCRM is widely studied as a means to **identify**, **assess**, **mitigate**, and **monitor** risks in supply chain operations (Gurtu and Johnny 2021; Ho et al. 2015; Sodhi, Son, and Tang 2012). These steps involve identifying potential

disruptions, assessing their likelihood and impact, mitigating risks through strategic planning and redundancy, and monitoring the supply chain continuously to ensure timely detection and response to emerging threats. Early studies focus on risk classification and mitigation strategies, whereas recent research applies artificial intelligence (AI), machine learning (ML), and blockchain to enhance proactive risk prediction and resilience (Baryannis et al. 2019; Emrouznejad, Abbasi, and Sıcakyüz 2023). Despite these advances, current approaches still struggle with the complexity and volatility of modern supply networks. Many rely heavily on historical data and static scenarios, which limits responsiveness to real-time disruptions (Ganesh and Kalpana 2022), while the lack of a unified framework to integrate diverse risk factors keeps practices fragmented and reactive (Bak 2018; Prakash, Soni, and Rathore 2017). These gaps highlight the need for more adaptive and intelligent SCRM methods, with recent advances in LLMs offering new opportunities for semantic reasoning and contextual decision support.

To address the limitations of traditional SCRM approaches, researchers have introduced methods such as predictive risk analytics based on ML (Choi 2021; Yang et al. 2023), real-time simulation through digital twins (Ivanov and Dolgui 2021; Ivanov et al. 2019),

and agent-based modeling for adaptive risk management (Bi et al. 2022; Blos et al. 2016). Among these, agent-based approaches are particularly notable as they enable decentralised decision-making, dynamic interaction, and adaptive behaviour within supply chains (Chen et al. 2013). Building on this foundation, MAS emerge as a promising paradigm for managing supply chain risks. Recent surveys show that MAS outperforms centralised systems in managing complex and dynamic risks (Hassan et al. 2018), enabling autonomous agents to coordinate for distributed sensing, real-time assessment, and adaptive response (Giannakis and Louis 2011, 2016). Applications of MAS in supply chain contexts include logistics coordination, maintenance risk analysis, and supply network agility enhancement (Gallab et al. 2017; Y.-L. Wang, Ye, and Yan 2010).

While MAS enhance SCRM, challenges persist in semantic understanding, agent coordination, and scalability. Limited ability to process unstructured data, communication complexity, and high computational costs constrain their effectiveness in large, dynamic networks (W. Shen et al. 2006; L. Xu, Mak, et al. 2024). To address these gaps, recent studies explore the integration of LLMs into MAS architectures (Quan and Liu 2024b; L. Xu, Almahri, et al. 2024). LLMs enhance agent capabilities through richer reasoning, adaptive dialogue, and better handling of unstructured data. This integration can greatly improve the responsiveness, adaptability, and resilience of MAS in SCRM, pointing to a promising direction for future research.

2.2. LLMs applications in supply chain management

Recent advancements in LLMs demonstrate transformative potential in supply chain management (SCM) and provide foundational insights with frameworks for their integration. Wang et al. (2024) show that LLMs can enhance decision-making, efficiency, and resilience in SCM while addressing security and ethical issues, whereas Hendriksen (2023) highlights their sociotechnical potential to transform supply chain practices.

Applications of LLMs in SCM span various aspects. OptiGuide uses LLMs for interpreting optimisation results and enabling natural language queries in supply chain decision-making (B. Li et al. 2023). InvAgent applies zero-shot learning for adaptive, cost-efficient inventory management (Quan and Liu 2024b), while CRMAgent adopts a multi-agent LLM framework to generate CRM templates via group learning and retrieval adaptation (Quan, Li, and Chen 2025). LLMs have also been used for supply chain network extraction and classification. T. Liu and Meidani (2024) show that fine-tuned LLMs reveal hidden relationships and roles,

improving the mapping of complex supply chain networks. Additionally, EconLogicQA and EconWebArena evaluate LLMs and agents on reasoning, grounded understanding, and real-world multimodal tasks in supply chain and economic settings (Z. Liu and Quan 2025; Quan and Liu 2024a). LLMs improve forecasting, inventory management, and logistics efficiency (Fosso Wamba et al. 2024; Khlie et al. 2024), and their integration with knowledge graphs enhances visibility, decision-making, and risk control across multi-tiered supply chains (AlMahri, Xu, and Brintrup 2024).

Risk management stands as a key research focus in the application of LLMs within SCM. Recent studies introduce frameworks such as CERIA for early disruption detection, risk categorisation from unstructured data, competency oriented decision models, and SHIELD for predictive analytics (Cheng et al. 2024; Jackson et al. 2024; Shahsavari et al. 2024; Zhao et al. 2024). Remarkably, LLMs have been leveraged to support risk assessment by combining structured information, such as operational and financial indicators, with unstructured sources like event reports and news, enabling more comprehensive and context-aware evaluation of supply chain risks (Quan et al. 2025). Although this prior study provides a foundation for our paper, we further introduce multi-agent iterative mechanisms to achieve more adaptive and collaborative risk management.

LLMs also enable multi-agent applications in SCM, opening new opportunities for intelligent interaction, coordination, and task execution. L. Xu, Almahri, et al. (2024) integrate LLMs into MAS to enhance decision-making in autonomous supply chains, while Zheng et al. (2024) show that LLM-based agents improve visibility, forecasting, and risk mitigation through real-time disruption detection. Despite these promising directions, the use of LLMs in MAS for SCRM remains under-explored. This gap motivates further exploration of LLM-based MAS, which we review in the following subsection.

2.3. LLM-based multi-agent systems

The emergence of LLMs enables new possibilities for building collaborative MAS. Unlike traditional agents with fixed rules or domain-specific protocols, LLM-based agents feature advanced language understanding, generation, and contextual reasoning, making them effective in multi-agent collaboration where flexibility and shared understanding are essential (Talebirad and Nadiri 2023). Recent surveys show that LLM-driven agents can act as planners, executors, or mediators in dynamic teams, providing a more human-like and adaptive form of distributed intelligence (Guo et al. 2024; X. Li et al. 2024). As a result, LLM-based multi-agent collaboration is

emerging as a promising paradigm for addressing complex, real-world challenges, including those in SCRM.

Recent work applies LLMs in MAS to support collaborative planning, perception, and decision-making, which align closely with coordination challenges in supply chain operations. Tran et al. (2025) outline coordination mechanisms such as task decomposition, role planning, and dialogue negotiation. Complementary efforts such as MetaGPT (Hong et al. 2023), CAMEL (G. Li et al. 2023), HuggingGPT (Y. Shen et al. 2023), AutoGen (Q. Wu et al. 2024), and AutoBnB (Z. Liu 2024, 2025) demonstrate LLMs collaboration in software development, tool orchestration, and role-playing scenarios. Together, these studies illustrate the growing maturity of LLM-based multi-agent collaboration and its potential to address the complex, distributed decision-making needs of SCM.

Our framework differs from prior multi-agent LLM systems in that MARS is designed as a domain-specific, risk-focused architecture rather than a general-purpose coordination framework. In contrast to existing approaches that emphasise collaborative planning or tool orchestration, MARS integrates an iterative feedback-revision loop, extracts quantitative risk signals from unstructured narratives, and combines them with structured supply chain indicators to support decision-making. This design enables the framework to produce risk-informed facility location assessments in a manner that extends beyond the capabilities of previous multi-agent LLM approaches.

Supply chains are complex, data-driven, and disruption-prone, demanding intelligent systems for dynamic coordination and contextual decision-making. Traditional SCRM struggles with fragmented data, delays, and rigid rule-based responses. MAS distribute decision-making across autonomous agents but remain limited by handling of unstructured data and inflexible collaboration. LLM-based MAS overcome these limits by combining distributed agents with advanced language understanding, adaptive reasoning, and semantic coordination. Building on these insights, we propose an LLM-enhanced multi-agent framework, MARS, that leverages the strengths of LLM-based MAS in semantic interpretation, adaptive reasoning, and distributed coordination, which we apply for risk-aware logistic hub site selection within supply chain networks.

3. Problem definition

Logistic hub site selection is a critical decision in supply chain networks, directly affecting operational efficiency, cost, and system resilience. Traditionally, such decisions rely on structured and quantitative indicators. However, as supply chains face increasing uncertainty

from environmental disruptions, there is growing recognition that valuable risk information also exists in unstructured textual sources, such as event reports or narrative descriptions. To support robust decision-making under uncertainty, it becomes essential to integrate both structured and unstructured risk data. The core challenge lies in the fusion of these heterogeneous risk dimensions. While structured indicators can be directly computed and normalised, unstructured narratives often contain context-dependent information that are difficult to interpret and quantify using traditional models. This highlights the need for intelligent systems capable of extracting, reasoning, and synthesising diverse forms of risk information to enable comprehensive and adaptive site evaluation.

Problem Description: Given a set of candidate logistic hub locations, each associated with structured risk attributes (e.g. cost, distance, traffic accidents, traffic congestion, and crime) and unstructured risk inputs (e.g. extreme weather event narratives), the goal is to determine a risk-aware ranking that reflects each site's operational suitability under uncertainty.

In summary, the problem of logistic hub site selection under uncertainty demands a comprehensive solution that goes beyond traditional structured indicators. Effectively addressing this challenge requires integrating heterogeneous risk information, including both quantitative metrics and unstructured narratives, to enable robust and contextual decision-making. This motivates the development of a unified framework, MARS, capable of fusing diverse data types for resilient and informed site evaluation, which we present in Section 5.

4. Data

To support risk-aware logistic hub site selection, we construct a testbed using target and candidate logistic hub locations from the southeastern United States to evaluate our proposed framework. Specifically, we aim to identify the most suitable logistic hub candidate from a set of realistic industrial sites as potential matches for pre-defined target locations. This requires defining two types of hubs: (1) target hubs, representing desired long-term infrastructure positions; and (2) candidate hubs, drawn from actual commercial property listings and serving as feasible options to realise those target hubs. Together, candidate hubs provide the search space from which suitable locations for target hubs are selected. Although our testbed focuses on the southeastern United States, the same data structure and workflow can be applied to other regions by replacing the geographic inputs with analogous datasets.

We identify target logistic hubs adapted from the designed hyperconnected logistic hub network. The hyperconnected network is designed through principles mentioned by Montreuil (2011), Kulkarni, Dahan, and Montreuil (2024), Y. Xu et al. (2024c), and Garbers, Muthukrishnan, and Montreuil (2025). These target hubs are not actual industrial properties, but rather represent major highway intersections or regional centres where suitable properties should be selected from the broader set of candidate hubs.

In contrast, logistic hub candidates are selected from existing real-world industrial properties listed in the CoStar commercial real estate database.¹ We apply a uniform filtering criterion across states to identify lease available industrial sites with 10,000 to 50,000 square feet of space, classified under ‘existing’ construction status. We also require a parking ratio of at least 2 spaces per 1000 square feet to ensure operational feasibility, including space for employees, vehicle circulation, and small delivery truck staging. This implies that approximately one-third of the total area would need to be allocated to parking and circulation functions. As of April 2025, numerous such listings are identified across key south-eastern U.S. states, specifically: 17 in Alabama, 162 in Florida, 102 in Georgia, 2 in Mississippi, 47 in North Carolina, 21 in South Carolina, and 25 in Tennessee. These candidate hubs represent realistic infrastructure that could potentially support future development of target hubs, as illustrated in Figure 2.

5. Methodology

This section outlines the methodology used in this study. We first present an overview of the proposed MARS framework, followed by a detailed description of the LLM-based extraction of unstructured risks. We then discuss additional structured risk factors and explain how these dimensions are integrated into a composite scoring scheme for ranking candidate logistic hubs.

5.1. Framework overview

To address RQ1, we propose a LLM-enhanced multi-agent framework, MARS, for conducting risk assessment in supply chain networks. MARS integrates both unstructured and structured risk factors, enabling iterative refinement of unstructured risks while systematically incorporating other structured dimensions. To process unstructured risk information, we leverage three specialised agents enhanced by LLM capabilities. *RiskAgent* leverages LLM-based semantic reasoning to transform qualitative textual evidence into quantitative severity levels. *FeedbackAgent* reviews the overall risk assessment results and provides feedback to enhance

their reliability and consistency. *RevisionAgent* then updates the initial severity estimates based on this feedback, creating an iterative improvement loop with *FeedbackAgent* that terminates either after a fixed number of iterations or once consecutive ratings converge. By integrating both quantitative and qualitative risk sources, the framework supports interpretable, data-driven decision-making under uncertainty.

As a case study to instantiate the proposed MARS framework, we focus on the logistic hub site selection in this paper. Candidate locations are evaluated across six key risk dimensions: cost, distance, traffic accidents, traffic congestion, crime, and extreme weather events. The overall process is illustrated in Figure 1, which shows how three agents interact through a feedback-revision loop to refine risk assessments. In this case study, unstructured risks are refined iteratively, whereas structured factors are systematically computed and normalised to ensure consistent evaluation across candidates. MARS is designed to accommodate diverse forms of unstructured risk information, ranging from community opposition and protest reports to context-specific disruptions. In our testbed, this capability is demonstrated using weather-related reports, which are iteratively refined to assess their potential impact on logistic hub site selection.

5.2. LLM-based unstructured risk assessment

To answer RQ2, we implement a multi-agent refinement process, as illustrated in Figure 3. This part of the workflow further details the process shown in Figure 1. In Step 1, *RiskAgent* converts event narratives into initial severity levels and rationales. In Step 2, *FeedbackAgent* evaluates this output across multiple dimensions, including severity accuracy, narrative fidelity, reason clarity, and overall confidence. Severity accuracy refers to how well the assigned score reflects the event details; narrative fidelity indicates how faithfully the assessment stays aligned with the original description; reason clarity describes how clear and logically sound the supporting explanation is; and overall confidence reflects the evaluator’s confidence in the validity of the severity output. Step 3 involves *RevisionAgent*, which revises the severity assessment based on the provided feedback. Finally, Step 4 allows for a second round of evaluation to assess the quality of the revised output relative to the original. This iterative feedback-revision loop enables systematic, LLM-driven refinement of unstructured risk information, forming the foundation of the agent workflow described in the subsections that follow.

To illustrate how the above workflow can be implemented in a concrete setting, we present its application to

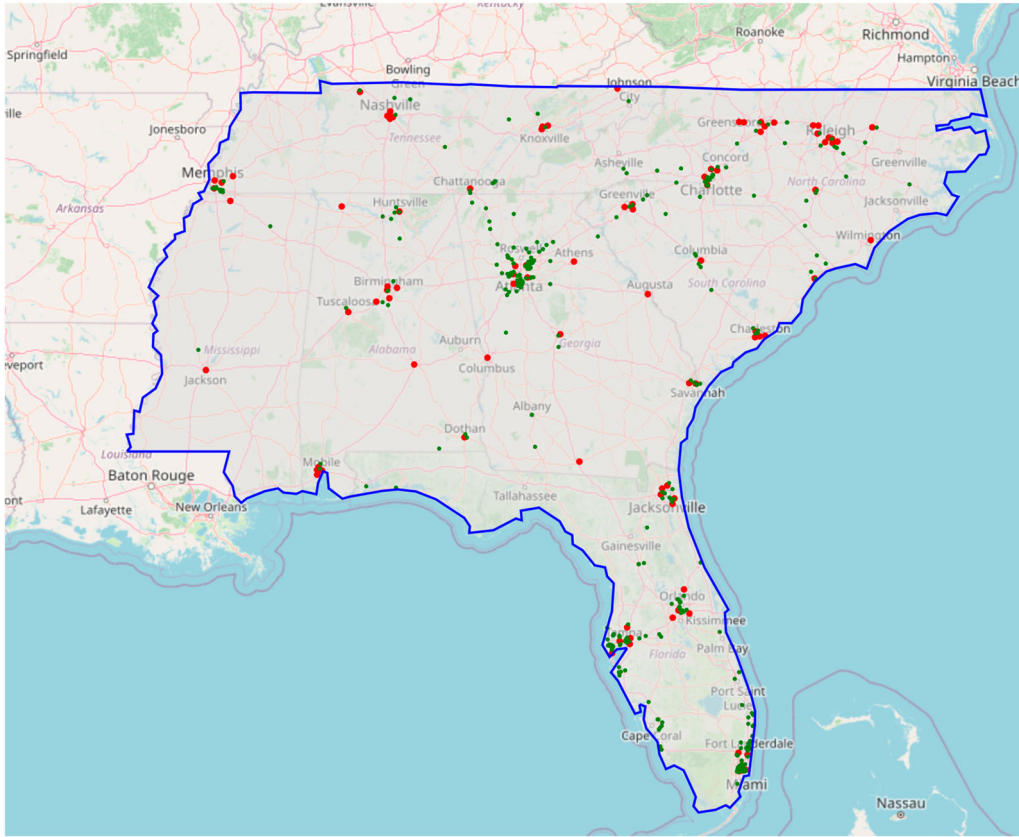


Figure 2. Distribution of target and candidate logistic hubs in the southeastern United States.

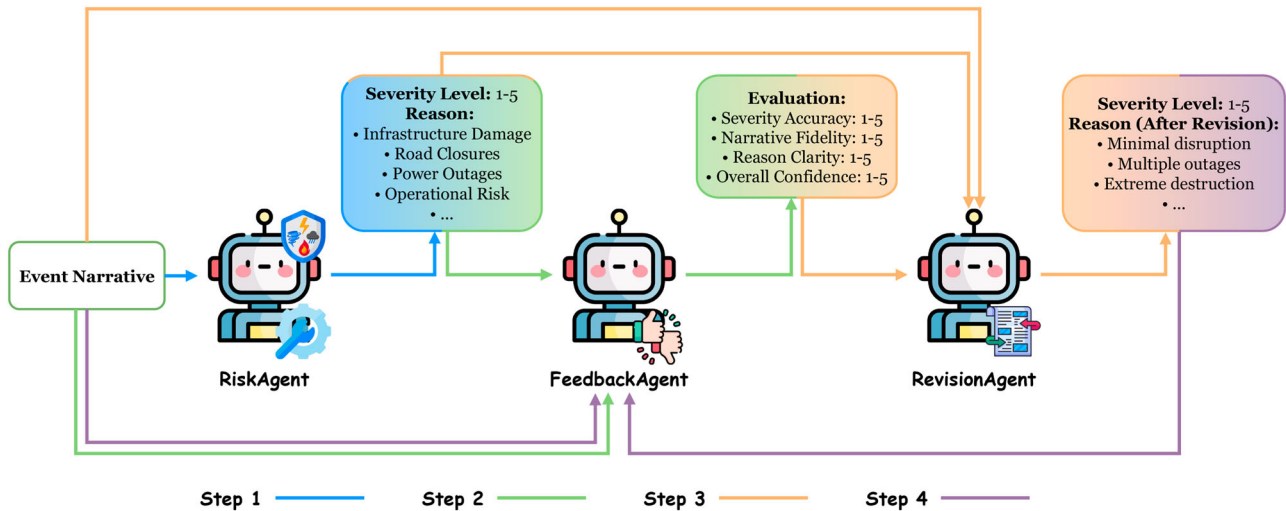


Figure 3. Multi-agent feedback-revision loop for LLM-based severe event risk assessment.

unstructured risk assessment in our testbed. Such unstructured risk information may include weather-related reports, community opposition or protest records, supplier incident reports, and other narrative sources that capture disruptions beyond structured quantitative indicators; in this paper, we focus on the weather-related case study for demonstration. Weather-related unstructured information constitutes a critical factor in evaluating the operational resilience of candidate hub locations under uncertainty. We assess severe weather risk at candidate hub locations using National

Oceanic and Atmospheric Administration (NOAA) Storm Events Database,² a comprehensive collection of severe weather reports maintained by National Centres for Environmental Information³ (NCEI) under NOAA. This dataset contains detailed records of significant meteorological events across the United States, including hurricanes, tornadoes, floods, thunderstorms, winter storms, and other extreme weather phenomena. While structured meteorological variables (e.g. temperature, humidity, precipitation) could in principle complement this analysis, they are not used here because such

measurements are often missing or inconsistent in NOAA records, making cross-event comparison unreliable. In contrast, narrative fields are consistently available and directly describe operational impacts most relevant to disruption-focused risk assessment.

This agent workflow is orchestrated by LangGraph,⁴ enabling modular and interpretable multi-agent execution. LangGraph provides a framework for composing and managing agent interactions as modular nodes within a directed workflow graph. Each agent is encapsulated as a node with clearly defined inputs and outputs, enabling flexible connection, monitoring, and extension. This modular structure supports transparent debugging and facilitates adding or replacing agents without the need to redesign the entire system.

5.2.1. RiskAgent

RiskAgent is a retrieval-based reasoning component designed to assess the severity of severe events and their potential impact on critical operations such as logistics, supply chains, or infrastructure. It operates by retrieving relevant event records, analyzing narrative descriptions, and mapping them to structured severity levels. By converting unstructured text into interpretable risk assessments, the agent enables consistent evaluation across diverse event types.

In our case, RiskAgent is specialised for weather-related severe events. For each candidate hub, the agent dynamically queries NOAA Storm Events Database to retrieve daily weather data, ensuring spatio-temporal alignment with hub-level evaluations. Leveraging the `NOAA_storm_event_narrative` field in the retrieved data, the agent uses a tailored prompt (see “Prompt for RiskAgent”) to convert textual event descriptions into structured severity scores. The prompt defines a five-level severity rubric, where scores from 1 to 5 correspond to increasing levels of disruption, ranging from mild disturbances to extreme events involving widespread damage or casualties. Specifically, the five-level rubric is defined as follows:

- **Level 1:** Minimal impact, no significant disruptions, minor weather effects (e.g. light rain or wind).
- **Level 2:** Low impact, localised disturbances, minor flooding, or moderate wind speeds causing slight delays.
- **Level 3:** Moderate impact, operational disruptions such as road closures, localised structural damage, or power outages.
- **Level 4:** High impact, significant disruptions, widespread infrastructure damage, major flooding, or strong winds causing logistical challenges.

- **Level 5:** Severe impact, extreme conditions (e.g. hurricanes or tornadoes), widespread destruction, or long-term operational shutdowns.

Prompt for RiskAgent

You are an assistant specialized in evaluating weather-related risks for logistic hubs.

Given the following logistic hub location and historical weather data, extract weather-related risks.

- Location:
 - Latitude: {latitude}
 - Longitude: {longitude}
 - County: {county}
 - State: {state}

- Date Range: {start_date} to {end_date}

Storm Data: {NOAA_storm_event_narrative}

Assign a severity level to each weather event on a scale from 1 to 5 based on the event narrative and overall impact:

- 1: Minimal impact, no significant disruptions, minor weather effects such as light rain or wind.
- 2: Low impact, localized disturbances, minor flooding, or moderate wind speeds causing slight delays.
- 3: Moderate impact, disruptions to operations, road closures, structural damage, or power outages in some areas.
- 4: High impact, significant disruptions, widespread infrastructure damage, major flooding, or high wind speeds causing logistical challenges.
- 5: Severe impact, extreme weather conditions such as hurricanes or tornadoes, widespread destruction, long-term operational shutdowns.

The model is instructed to output both a numerical score and a brief justification, referencing impact descriptors and context cues present in the narrative. An example output is shown in “GPT-4o Response of RiskAgent”, where severe weather event types and their corresponding severity levels are extracted for a given location and time. This process enables interpretable, location-specific risk quantification that feeds directly into logistic hub evaluation and selection. Note that the event ID shown in the example comes directly from the NOAA storm event record included in the retrieved storm data. Although the prompt does not explicitly mention an event ID field, the NOAA record provided to the model contains this information, and the model simply reflects it in its output.

GPT-4o Response of RiskAgent

Latitude: 34.681486
 Longitude: -84.488994
 County: Gilmer
 State: GA
 Date: 2023-06-25
 Risks:
 Event ID: 1109432
 Event Type: Thunderstorm Wind
 Event Narrative: Emergency manager reported a tree down blocking one lane on Tails Creek Rd near the intersection with Pleasant Grove Rd.
 Severity Level: 3
 Reason: Multiple incidents of trees down blocking roads, causing significant disruptions.

In our experimentation, RiskAgent processes 11,858 extreme weather event narratives from the year 2023, retrieved from the NOAA Storm Events

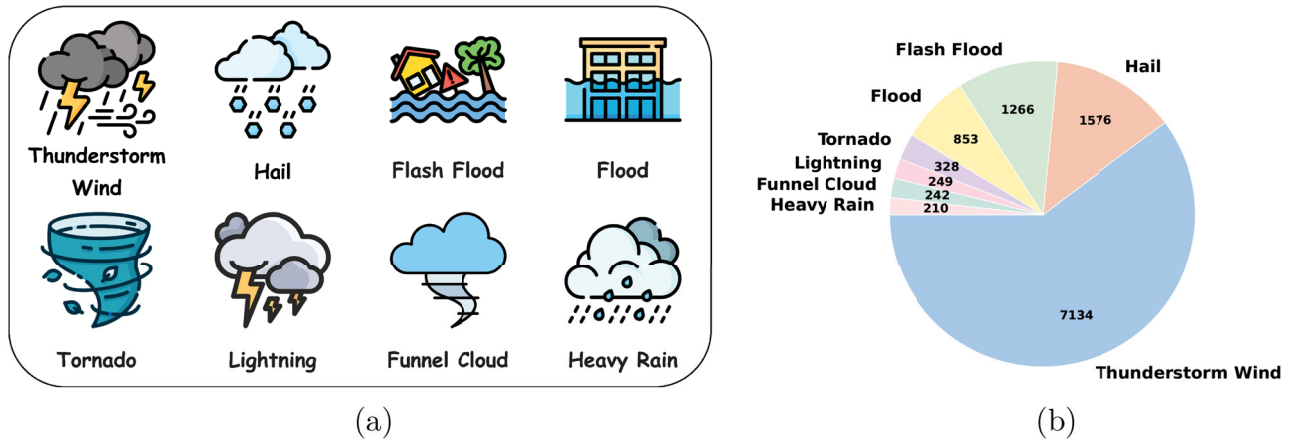


Figure 4. Severe weather categories and their distribution across candidate logistic hubs based on RiskAgent analysis. (a) Severe weather event categories identified by RiskAgent and (b) Distribution of severe weather categories across candidate hubs.

Database, across 376 candidate logistic hub locations using LLM-based semantic reasoning, focussing on a single year to ensure temporal consistency and to align with the most recent complete dataset. RiskAgent identifies eight primary categories of severe weather, including thunderstorm wind, hail, flash flood, flood, tornado, lightning, funnel cloud, and heavy rain, with the category types illustrated in Figure 4(a) and their corresponding counts summarised in Figure 4(b). The geographic distribution of these weather events reveals substantial variation across southeastern U.S. states, as shown in Table 1. Florida reports the highest frequency across most categories, including flood (828), lightning (182), and tornado (188), likely reflecting its large coastal exposure and high population density. In contrast, Mississippi exhibits far fewer reported events overall, with only 19 cases of thunderstorm wind and fewer than 15 events in any other category. Georgia and Alabama show relatively high counts for thunderstorm wind (2,953 and 538 respectively), with Georgia considerably higher, yet both states stand out relative to other regions. While our analysis centres on a single year to maintain temporal consistency, the pipeline itself is fully capable of integrating multi-year or multi-source unstructured risk data in the same manner.

Each severe weather event identified by RiskAgent is assigned a severity level, following a structured rubric that ensures consistency across narratives and locations. This enables clear comparison of event impacts, forming the basis for the subsequent severity distribution analysis. The resulting severity distribution is moderately skewed toward the lower end: 683 events receive a score of 1, 5015 are rated 2, and 5454 are rated 3. In contrast, 630 events are assigned a score of 4, and just 76 receive the maximum severity level of 5. This distribution indicates that most extreme weather

events exhibit mild to moderate impacts. However, the presence of high severity events, particularly levels 4 and 5, suggests the need to account for rare but consequential disruptions in risk-aware hub evaluation. These events, though infrequent, may disproportionately affect operational resilience and therefore warrant careful attention in downstream planning. While RiskAgent enables broad and granular coverage of extreme weather event types, further refinement is needed to ensure consistency and reliability in the generated severity assessments. These tasks are subsequently handled by FeedbackAgent.

5.2.2. FeedbackAgent

FeedbackAgent is a quality assurance component designed to evaluate the reliability and interpretability of outputs generated by other agents. Its core role is to provide structured, multi-dimensional feedback on risk assessments, ensuring that results are not only numerically consistent but also logically and contextually sound. By systematically scoring outputs across well-defined dimensions and providing concise justifications, FeedbackAgent enables transparent evaluation, error detection, and iterative refinement. Specifically, the assessment focuses on the following dimensions:

- **Severity Accuracy:** The extent to which the assigned severity score accurately reflects the details described in the event narrative. (1 = Completely inaccurate, 5 = Completely accurate)
- **Narrative Fidelity:** The degree to which the severity assessment remains faithful to the original event description without introducing external assumptions. (1 = Highly distorted or speculative, 5 = Fully faithful to the narrative)

- **Reason Clarity:** The clarity, coherence, and logical soundness of the provided explanation (reason) supporting the severity assignment. (1 = Unclear or incoherent, 5 = Very clear and logically sound)
- **Overall Confidence:** The overall confidence level in the validity of the severity output and its justification. (1 = No confidence, 5 = Full confidence)

Notice: The overall confidence score reflects a holistic judgment of the severity output and its justification, considering factors beyond the individual criteria, such as the evaluator’s overall trust in the assessment and consistency with domain knowledge.

In our case, for each identified weather event, `FeedbackAgent` takes as input `NOAA_storm_event_narrative` retrieved from NOAA Storm Events Database, along with the assigned severity level and supporting reason generated by `RiskAgent`. `FeedbackAgent` conducts a systematic evaluation of output quality across four assessment dimensions. Each of these dimensions is scored individually, and concise explanations are provided to support the evaluation. To guide the evaluation process, we design a well-defined prompt that instructs the model to assess each output according to the specified dimensions and return a concise and structured response. The full prompt is shown in “Prompt for `FeedbackAgent`”. The prompt defines four evaluation dimensions in the above list. It also instructs the evaluator to rely solely on the event description and the accompanying reasoning, without introducing any external knowledge. An example response generated by `FeedbackAgent` is shown in “GPT-4o Response of `FeedbackAgent`”. The response includes an organised severity evaluation, dimension-wise scores, and concise justifications, enabling transparent feedback and downstream refinement.

Prompt for `FeedbackAgent`

You are an assistant specialized in evaluating the quality of weather-related risk severity outputs.

Given:

- A weather event description: `{NOAA_storm_event_narrative}`
- A severity score (1 to 5): `{severity_level}`
- A textual explanation (reason) supporting the severity assessment: `{reason}`

Please assess the output based on the following four dimensions:

1. Severity Accuracy: How accurately does the assigned severity reflect the actual event described?
2. Narrative Fidelity: Does the severity assessment remain faithful to the facts and impacts described in the narrative?
3. Reason Clarity: Is the provided explanation clear, coherent, and logically structured?
4. Overall Confidence: How confident are you in trusting the severity output and its supporting reason?

Evaluation guidelines:

- Base your assessment strictly on the provided event narrative and reason; do not introduce external knowledge.
- Maintain concise and structured feedback.

GPT-4o Response of `FeedbackAgent`

Event ID: 1109432

Event Type: Thunderstorm Wind

Event Narrative: Emergency manager reported a tree down blocking one lane on Tails Creek Rd near the intersection with Pleasant Grove Rd.

Severity: 3

Reason: Multiple incidents of trees down blocking roads, causing significant disruptions.

Evaluation:

Severity Accuracy: 2

Narrative Fidelity: 2

Reason Clarity: 2

Overall Confidence: 2

Comments:

Severity Accuracy: The severity score of 3 seems high for a single reported incident of a tree down blocking one lane, given it does not prevent road access fully or imply widespread damage.

Narrative Fidelity: The narrative only specifies a single tree down affecting one lane. The reason suggests multiple incidents and significant disruptions, which is not faithful to the narrative.

Reason Clarity: The reasoning does not clearly support the severity score, as it mentions multiple incidents not reflected in the provided narrative.

Overall Confidence: The mismatch between the narrative and the reasoning, along with the exaggerated severity score, reduces confidence in the assessment.

In summary, `FeedbackAgent` provides an interpretable mechanism for verifying the quality of severity assessments. By scoring each dimension individually and supplying concise justifications, it ensures that the outputs from `RiskAgent` are not only quantitatively evaluated but also qualitatively explained, forming a reliable basis for subsequent refinement and decision-making.

5.2.3. `RevisionAgent`

While `FeedbackAgent` provides comprehensive evaluations of severity assessments, the process remains incomplete without the ability to act on this feedback. To address this, we introduce `RevisionAgent`, which takes the initial assessment, event description, and detailed evaluation scores as input, and produces a revised severity level and justification when necessary. This step is essential for improving overall quality, especially in borderline or ambiguous cases where the initial severity assignment may be inaccurate or weakly justified. `RevisionAgent` operates by reviewing four core evaluation dimensions: severity accuracy, narrative fidelity, reason clarity, and overall confidence, and selectively revises the severity or explanation when any evaluation score falls below a set threshold (e.g., 4), ensuring the output better reflects the event described. Revisions are strictly constrained by the content of the original event narrative and supporting reason to prevent hallucination or over-interpretation. The full instruction prompt used to guide `RevisionAgent` is shown in “Prompt for `RevisionAgent`”.

Table 1. Count of severe weather event types in each southeastern U.S. state.

State	Flash flood	Flood	Funnel cloud	Hail	Heavy rain	Lightning	Thunderstorm wind	Tornado
Alabama	113	3	4	215	10	8	538	38
Florida	312	828	231	631	171	182	1559	188
Georgia	428	1	0	268	0	9	2953	34
Mississippi	1	0	0	12	0	1	19	6
North Carolina	161	14	2	179	20	18	807	27
South Carolina	94	6	4	86	9	26	419	25
Tennessee	157	1	1	185	0	5	839	10

Prompt for RevisionAgent

You are a revision assistant. Your job is to revise the severity score and the supporting reason if needed, based on evaluation feedback.

Original Event:

- Event Type: {event_type}
- Event Narrative: {original_narrative}
- Severity Level: {original_severity}
- Reason: {reason}

Evaluation Feedback: {evaluation}

Instructions:

1. Review the evaluation feedback across all four dimensions: severity accuracy, narrative fidelity, reason clarity, and overall confidence.
2. If **severity_accuracy** or **overall_confidence** is below 4, reconsider the assigned severity level and adjust it to better reflect the event described.
3. If **reason_clarity** or **narrative_fidelity** is below 4, rewrite the reason to improve clarity, structure, and alignment with the severity.
4. Do not introduce new information not present in the original input.
5. Ensure the revised reason faithfully supports the severity level.

An example output from RevisionAgent is presented in “GPT-4o Response of Revision Agent”. In this case, the original severity level of 3 received low evaluation scores from FeedbackAgent in the first turn, with a severity accuracy of 2, narrative fidelity of 2, reason clarity of 2, and overall confidence of 2, indicating that the initial assessment overstated the scale of the incident and included details (multiple incidents) not supported by the narrative. RevisionAgent revises the severity downward to 1 and rewrites the reason to accurately reflect the actual conditions, emphasising that a single tree down blocking one lane causes only minimal disruption. To assess the impact of this revision, the updated output is passed back into FeedbackAgent for re-evaluation. FeedbackAgent responds in the second turn with perfect scores of 5 across all dimensions, confirming that the explanation is fully consistent with the narrative and appropriately calibrated to the event’s true severity (see “GPT-4o Second-Turn Response of FeedbackAgent”). This example illustrates how our feedback-revision loop system enables targeted correction of weak outputs, enhancing calibration and interpretability in borderline or ambiguous cases. More representative examples of event narratives, their assigned

severity levels, and corresponding reasoning after revision can be seen in Appendix 2 Table A1.

GPT-4o Response of RevisionAgent

Event ID: 1109432
 Event Type: Thunderstorm Wind
 Original Severity: 3
 Revised Severity: 1
 Reason Before: Multiple incidents of trees down blocking roads, causing significant disruptions.
 Reason After: A single tree down blocking one lane does not cause major disruptions, justifying a level 1 severity.

GPT-4o Second-Turn Response of FeedbackAgent

Event ID: 1109432
 Event Type: Thunderstorm Wind
 Event Narrative: Emergency manager reported a tree down blocking one lane on Tails Creek Rd near the intersection with Pleasant Grove Rd.
 Severity Level: 1
 Reason: A single tree down blocking one lane does not cause major disruptions, justifying a level 1 severity.

Evaluation:

- Severity Accuracy: 5
- Narrative Fidelity: 5
- Reason Clarity: 5
- Overall Confidence: 5

Comments:

- Severity Accuracy: The severity score of 1 accurately reflects the minimal impact of the event described as a single tree blocking one lane.
- Narrative Fidelity: The assessment remains true to the narrative, accurately reflecting the minor nature of the incident.
- Reason Clarity: The explanation is clear, straightforward and logically supports the assigned severity score.
- Overall Confidence: High confidence is justified based on the correctness of the severity score and the clarity of the reasoning.

5.3. Structured risk extraction

Other risk factors relevant to logistic hub site selection are assessed using a diverse set of structured datasets. These include publicly available sources such as the US Accidents Dataset (Moosavi, Samavatian, Parthasarathy and Ramnath 2019; Moosavi, Samavatian, Parthasarathy, Teodorescu, et al. 2019), the National Performance Management Research Data Set (NPMRDS),⁵ and the National Incident-Based Reporting System,⁶ as well as proprietary records from the CoStar commercial real estate database. Each dataset captures a specific dimension of risk related to operational performance, including traffic accidents, traffic congestion, crime

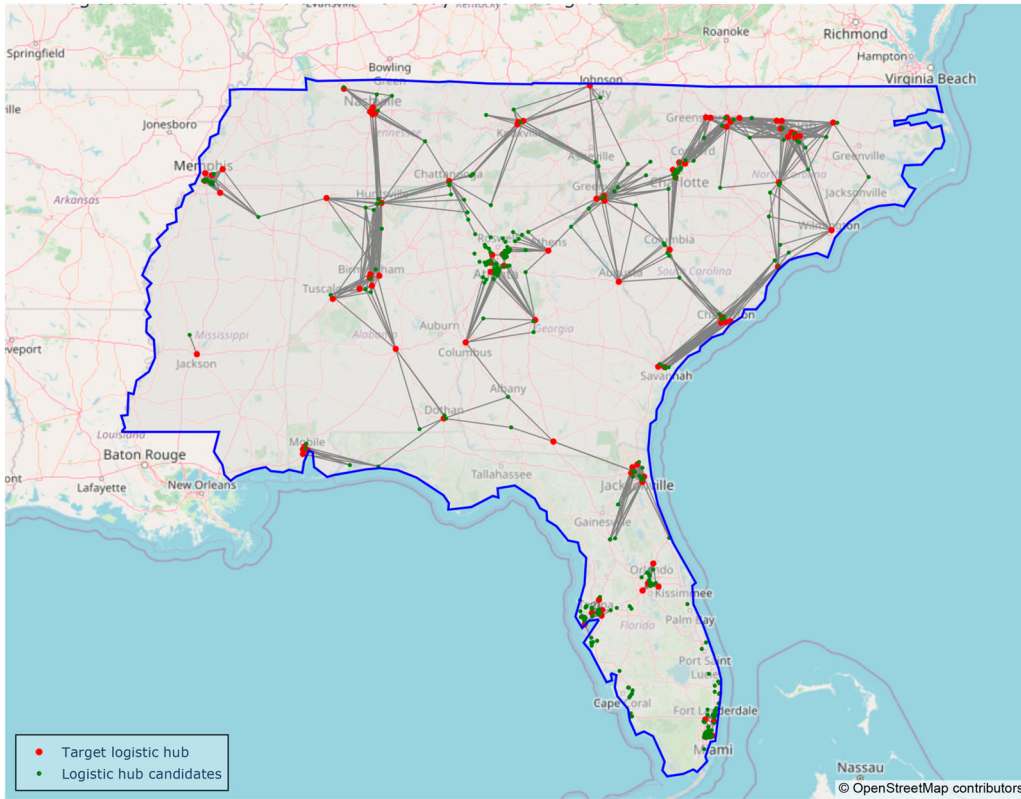


Figure 5. Assignment of candidate logistic hubs to target hubs based on distance in the southeastern United States.

rates, geographic distance, and facility annual rental cost.

5.3.1. Distance

Candidate logistic hubs are linked to target hubs based on geographic proximity, measured using Haversine distance derived from latitude and longitude coordinates (Sinnott 1984). For each target hub, up to c candidate sites within a d -mile radius are selected as potential matches. If fewer than c candidates fall within this threshold, all available candidates are retained. The overall spatial allocation is illustrated in Figure 5, where gray lines connect each target hub to its associated candidate sites. This distance-based filtering ensures that only geographically feasible locations are included in other risk evaluations. (In this research, we use $c = 15$ and $d = 100$.)

5.3.2. Traffic accidents

We use the US Accidents Dataset (2016–2023) to identify historical traffic accidents associated with each logistic hub candidate. Accident severity is categorised into four levels, from level 1 (minor) to level 4 (severe), and is used to evaluate the accident exposure within

each hub’s surrounding area. For each hub candidate, we define a circular catchment area with radius d , computed as $d = \sqrt{A_{\text{state}}/N}$, where A_{state} is the area of the state and N is the number of candidate hubs within that state. This formulation balances geographic scale with candidate density by ensuring that the total coverage across all hubs approximates the state’s area. States with fewer candidates yield larger catchment radii, while more densely covered states assign smaller search areas per hub.

To compute accident risk exposure, we extract all accidents located within each hub’s circular catchment area using spatial indexing. Specifically, we use the `BallTree`⁷ structure from the `scikit-learn` (Pedregosa et al. 2011) library, combined with the Haversine distance metric, to efficiently retrieve accident records within a distance d of each hub. The matched accidents are then grouped by day-of-year and severity level, and the average number of daily accidents of each severity is computed over the 8-year period following the previous work (Moosavi, Samavatian, Parthasarathy and Ramnath 2019; Moosavi, Samavatian, Parthasarathy, Teodorescu, et al. 2019) and dataset setting. These average counts are normalised to yield daily probabilities, denoted as $P_{\text{daily},h}(S = s)$.

5.3.3. Traffic congestion

Traffic congestion refers to a state of reduced vehicular speeds and increased travel times due to demand exceeding roadway capacity. In this study, congestion levels are identified using probe vehicle data extracted from NPMRDS. Following the methodology outlined in Federal Highway Administration's (FHWA) congestion and reliability guidance (Margiotta et al. 2018), congestion severity is classified based on the speed reduction factor (SRF), defined as the ratio of the average peak period speed to the free-flow speed:

$$\text{SRF} = \frac{v_t}{v_{\text{free}}} \quad (1)$$

where v_t denotes the observed hourly average speed, and v_{free} denotes the free-flow speed, defined as the mean speed observed during overnight hours (10:00 PM–5:00 AM) on weekdays. Based on the SRF value, congestion levels are categorised into three classes. According to the principles mentioned in FHWA, for freeway segments, SRF values between 90% and 100% indicate no to low congestion, values between 75% and 90% indicate moderate congestion, and values below 75% indicate severe congestion. For non-freeway segments, the thresholds are 80%–100% for no to low congestion, 65%–80% for moderate congestion, and below 65% for severe congestion. To associate congestion patterns with freight operations, each logistic hub candidate is linked to a set of nearby Traffic Message Channel (TMC) segments. The selected segments primarily include interstate highways and principal arterial roads, with a small portion consisting of minor arterials and major collectors. This method enables a spatially focused assessment of congestion severity around freight facility locations.

5.3.4. Crime

Crime is an essential component in comprehensive risk assessment. We obtain crime data from FBI National Incident-Based Reporting System (NIBRS), which provides incident information on offenses reported to law enforcement agencies across the United States. To assess the severity of different crime types, we reference the methodology of the Crime Severity Index (CSI) developed by Statistics Canada.⁸ Specifically, we take into consideration the following significant crime categories: murder and nonnegligent manslaughter, aggravated assault, robbery, burglary/breaking and entering, weapon law violations, and motor vehicle theft. These offenses are particularly relevant to logistic hub site selection due to their direct implications for employee safety, asset security, and operational risk. We apply a modified version of the CSI to quantify the overall crime severity

for each location, using the following formula:

$$\text{CSI} = \left(\frac{\sum (C_i \times W_i)}{P} \right) \times 100 \quad (2)$$

where C_i denotes the number of incidents for offense i , W_i is the severity weight assigned to that offense, and P represents the total population of the jurisdiction.

5.3.5. Cost

The annual rental cost of each candidate logistic hub is estimated using data from the CoStar property database. For each location, we multiply the reported annual rent per square foot (in U.S. dollars) by the total available space, yielding a standardised measure of yearly rental expenditure. This approach ensures consistency across all candidate hubs, providing a uniform basis for cost-aware evaluation. Importantly, in MARS cost is not treated as a source of variability or uncertainty, but rather as a relatively stable and structural factor that must be balanced against other dynamic risks. Whereas severe weather, traffic congestion, or traffic accidents may disrupt operations unpredictably, facility rental cost represents a long-term financial commitment that directly affects operational resilience. High-rent locations may provide access to better infrastructure or lower exposure to environmental risks, but they also increase fixed expenditures; conversely, low-cost sites may reduce financial burden while exposing operations to higher levels of disruption.

5.4. Risk aggregation for composite scoring

To solve RQ3, we perform risk aggregation to enable consistent and risk-informed ranking of candidate logistic hubs, allowing us to identify the most suitable hub for each target location. We take six types of risk into consideration, as shown in Figure 6. Among these, severe weather, traffic accidents, and traffic congestion have daily severity probability distributions, while distance, cost, and crime are relatively static or infrequently updated indicators.

For risk types with daily severity probabilities, we convert the daily metric into a yearly severity metric so that all risk dimensions share the same temporal resolution. The annual probability of observing at least one accident of severity s near hub h is estimated using the independent formulation given in Equation (3). This formulation integrates the daily accident probabilities $P_{\text{daily},h}$ over an entire year under the assumption of temporal independence, thereby yielding the annual probability $P_{\text{yearly},h}$ of

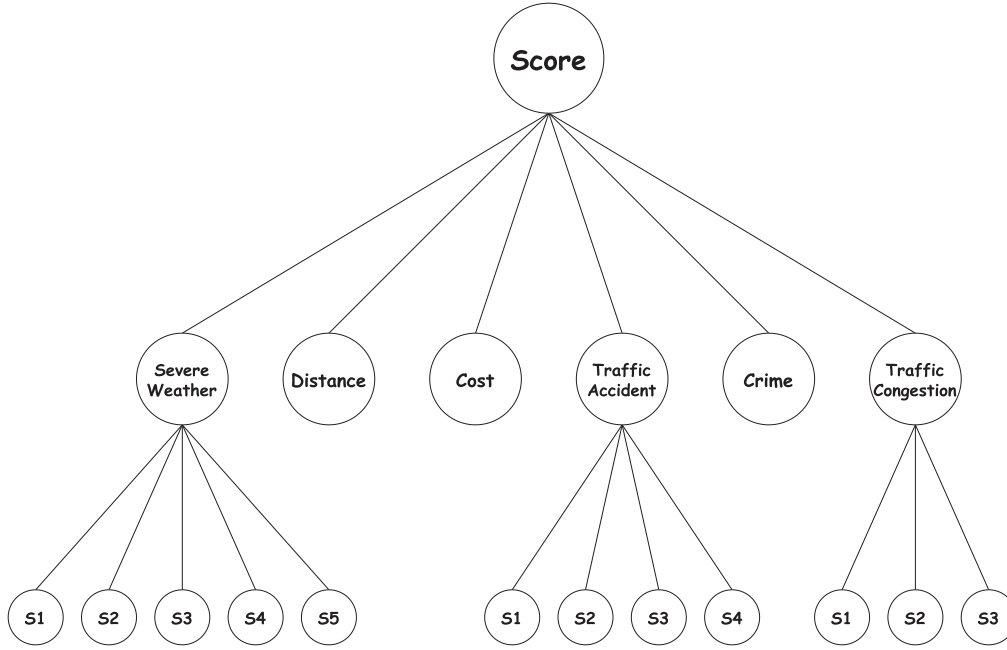


Figure 6. Hierarchical structure of the composite risk score. Multiple severity levels are first aggregated into sub-scores for each risk type, which are then combined with distance, cost, and crime indicators.

observing at least one accident of severity s near hub h .

$$P_{\text{yearly},h}(S = s) = 1 - \prod_{t=1}^{365} (1 - P_{\text{daily},h}(S = s)). \quad (3)$$

After we obtain the yearly probability for each severity level, we aggregate these probabilities into a single severity score for the corresponding risk type. Various aggregation strategies can be applied in this step, like simple averaging, maximum severity selection, weighted sums, weighted root mean square, and distance-based methods. In MARS, we adopt the distance-to-ideal-point approach described by Bellepeau et al. (2024), which measures how far each candidate deviates from an ideal profile (target logistic hub). To enable this comparison, probability-driven risk scores, distance, cost, and crime are first normalised to a common scale. The distance-to-ideal measure then accounts for the relative impact of different severity levels by applying weighted contributions. The resulting score provides a single, consistent indicator that integrates both risk and balancing dimensions, which are then jointly used in the composite scoring framework. Further implementation details are provided in Appendix 1.

6. Experiments

This section presents the results of the experimental investigation of the proposed MARS framework. We first

report the results of the unstructured risk extraction process, assessing the quality and consistency of the LLM-driven outputs. We also conduct ablation and sensitivity analyses to examine the effects of different agent configurations and design choices. We then provide an illustrative example showing how the extracted unstructured and structured risk factors are combined to score and rank candidate logistic hubs.

6.1. Experimental results of unstructured risk extraction

We assess the effectiveness of the LLM-driven extraction process by evaluating the performance of RiskAgent before and after applying the multi-agent feedback-revision loop. Tables 2 and 3 summarise the comparative results across four key evaluation dimensions: severity accuracy, narrative fidelity, reason clarity, and overall confidence, across eight categories of severe weather events. The event IDs used here are deduplicated to avoid multiple records of the same severe weather type being counted repeatedly. This ensures that each event is uniquely represented, preventing bias and yielding a more accurate evaluation of the LLMs' performance. By default, we use GPT-4o as the backbone model for all three agents (RiskAgent, FeedbackAgent, and RevisionAgent).

From Table 2, we observe that using FeedbackAgent to evaluate RiskAgent already yields reasonably good results before revision, and the incorporation

Table 2. Performance of weather severity extraction on severity accuracy and narrative fidelity across different event categories, with and without revision.

Category	Severity accuracy			Narrative fidelity		
	w/o Rev.	w/ Rev.	Δ (%)	w/o Rev.	w/ Rev.	Δ (%)
Thunderstorm wind	4.01	4.12	2.74%	4.60	4.68	1.74%
Hail	4.21	4.37	3.80%	4.74	4.83	1.90%
Flash flood	4.02	4.09	1.74%	4.47	4.69	4.92%
Flood	4.08	4.28	4.90%	4.67	4.83	3.43%
Tornado	3.91	4.01	2.56%	4.54	4.78	5.29%
Lightning	4.02	4.10	1.99%	4.54	4.75	4.63%
Funnel cloud	3.93	3.98	1.27%	4.63	4.66	0.65%
Heavy rain	4.68	4.54	-2.99%	4.96	4.86	-2.02%
Overall	4.05	4.15	2.47%	4.61	4.72	2.39%

Table 3. Performance of weather severity extraction on reason clarity and overall confidence across different event categories, with and without revision.

Category	Reason clarity			Overall confidence		
	w/o Rev.	w/ Rev.	Δ (%)	w/o Rev.	w/ Rev.	Δ (%)
Thunderstorm wind	4.16	4.33	4.09%	4.04	4.14	2.48%
Hail	4.34	4.48	3.23%	4.22	4.38	3.79%
Flash flood	4.10	4.22	2.93%	4.04	4.11	1.73%
Flood	4.13	4.23	2.42%	4.12	4.31	4.61%
Tornado	4.07	4.12	1.23%	3.94	4.03	2.28%
Lightning	4.14	4.22	1.93%	4.02	4.13	2.74%
Funnel cloud	3.95	4.12	4.30%	3.93	4.05	3.09%
Heavy rain	4.75	4.68	-1.47%	4.68	4.54	-2.99%
Overall	4.17	4.33	3.84%	4.07	4.17	2.46%

of *RevisionAgent* brings further improvements. Table 2 presents the performance on two key dimensions: severity accuracy and narrative fidelity. Severity accuracy measures how well the assigned severity level matches the actual event described, reflecting the model's ability to capture the true magnitude of the hazard. Narrative fidelity evaluates whether the severity assessment remains faithful to the facts and impacts detailed in the narrative, indicating how consistently the model grounds its judgment in the underlying textual evidence. Both show consistent gains once the revision process is applied. For example, in the case of flood events, severity accuracy improves from 4.08 to 4.28 (+4.90%), while narrative fidelity rises from 4.67 to 4.83 (+3.43%). Similar upward trends are observed for tornado and flash flood events. The only exception is heavy rain, where revision slightly reduces scores, likely because such events are often less distinctive in their reported impacts and harder to differentiate in severity, which makes interpretation more difficult.

From Table 3, we observe that adding *RevisionAgent* improves performance on reason clarity and overall confidence for most categories. Reason clarity measures whether the explanation is clear, coherent, and logically structured, reflecting the model's ability to articulate its reasoning in a way that is intelligible to humans. Overall confidence evaluates how trustworthy the severity output and its supporting explanation

appear. Representative gains are evident across several categories. For instance, in the case of hail events, reason clarity improves from 4.34 to 4.48 (+3.23%), while overall confidence rises from 4.22 to 4.38 (+3.79%). Flood events also show clear benefits, with overall confidence increasing from 4.12 to 4.31 (+4.61%), indicating that revisions made the outputs more trustworthy. Funnel cloud events illustrate the impact on explanations most strongly, where reason clarity improves from 3.95 to 4.12 (+4.30%). These examples highlight how the revision process not only refines the clarity of reasoning but also bolsters users' confidence in the severity assessments across diverse event types. In summary, the revision step strengthens how well the model explains itself and how dependable those explanations are judged to be, complementing the accuracy and fidelity gains reported in Table 2.

Overall, these results demonstrate that the revision process enhances the reliability (severity accuracy and overall confidence) and interpretability (narrative fidelity and reason clarity) of severity assessments, enabling the model to provide more accurate and faithful representations of extreme weather events. In addition, the findings show that the multi-agent feedback-revision loop provides consistent improvements in both accuracy and interpretability of severity extraction. These changes reflect relative differences between assessments with and without revision, and the recurring positive

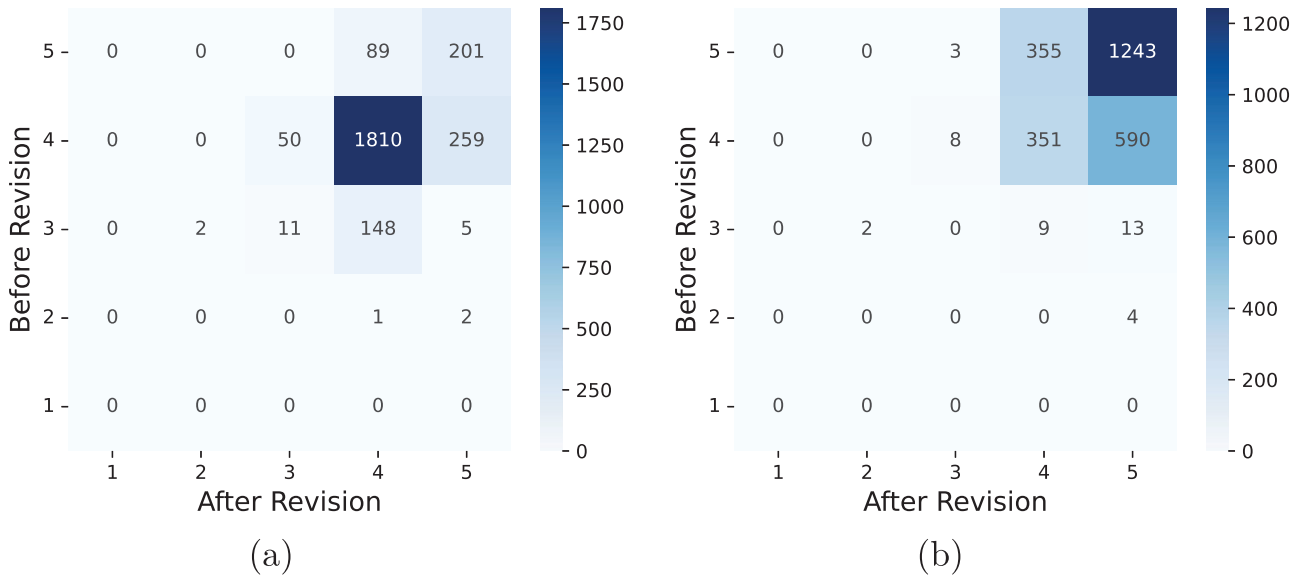


Figure 7. Distribution of evaluation scores before and after revision for (a) severity accuracy and (b) narrative fidelity, with diagonal concentration indicating consistency.

shifts across most categories highlight the robustness of the process rather than the magnitude of individual values. By simultaneously improving the factual alignment of severity judgments with event narratives and refining the clarity and trustworthiness of the accompanying explanations, the framework delivers assessments that are not only more precise but also more transparent and credible. This dual improvement highlights the value of integrating a feedback-revision loop into LLM-based extraction pipelines for high-stakes domains such as extreme weather analysis.

The heatmaps in Figures 7 and 8 illustrate how evaluation scores shift before and after revision across the four dimensions of severity accuracy, narrative fidelity, reason clarity, and overall confidence. In Figure 7, both severity accuracy and narrative fidelity show a clear concentration along the diagonal, confirming that most judgments remain consistent after revision. Importantly, upward shifts toward the top-right corner are frequent, reflecting that revision not only preserves prior high-quality assessments but also improves weaker cases by moving them to higher score levels. This explains the overall performance gains reported in Table 2.

Figure 8 presents a similar trend for reason clarity and overall confidence. The diagonal concentration again indicates stability, while the off-diagonal movements demonstrate notable improvements. In particular, many cases originally rated 4 are shifted to 5 after revision, especially for overall confidence, suggesting that the revision step makes severity assessments more trustworthy and their supporting explanations more persuasive.

These distributions confirm that revision plays a dual role: maintaining reliability where initial outputs are already strong and elevating weaker responses to higher quality.

6.2. Result analyses

This section analyzes the performance of the proposed MARS framework using one month of NOAA Storm Events database collected in May 2023. The analysis focuses on evaluating how different agent configurations, backbone models, and iterative mechanisms affect the quality, consistency, and stability of the evaluation results.

6.2.1. Model configurations

Table 4 compares different `RiskAgent` configurations GPT-4o,⁹ o4-mini,¹⁰ and GPT-5 mini,¹¹ while keeping `FeedbackAgent` and `RevisionAgent` fixed as GPT-4o. The purpose of this setup is to investigate how different `RiskAgent` configurations influence evaluation outcomes, while keeping `FeedbackAgent` and `RevisionAgent` fixed to prevent potential bias that could occur if identical model configurations are used for both assessment and generation. This design ensures that any observed differences can be attributed to `RiskAgent` itself rather than confounding effects from overlapping configurations.

Results in Table 4 demonstrate that the MARS framework exhibits strong compatibility and consistency across different LLM backbones. Despite using models of varying capacity and scale (GPT-4o, o4 mini, and GPT-5

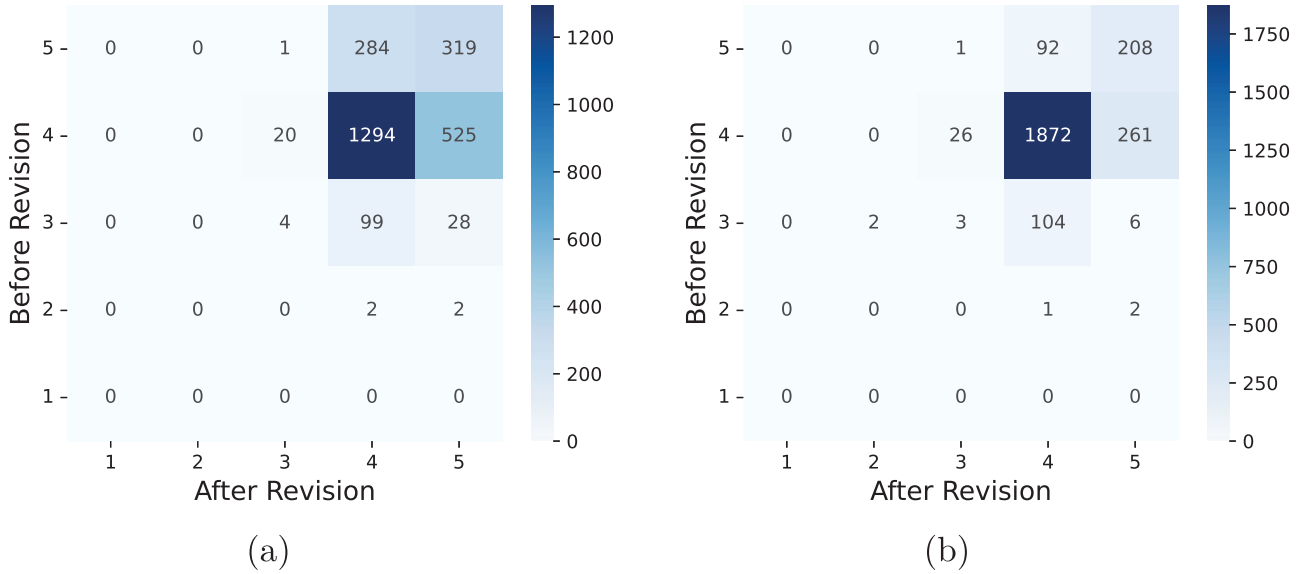


Figure 8. Distribution of evaluation scores before and after revision for (a) reason clarity and (b) overall confidence, with diagonal concentration indicating agreement. (a) Reason clarity and (b) Overall confidence.

Table 4. Performance comparison of severity accuracy, narrative fidelity, reason clarity, and overall confidence of various model configurations for RiskAgent.

Risk	Feedback	Revision	Severity accuracy	Narrative fidelity	Reason clarity	Overall confidence
GPT-4o	GPT-4o	GPT-4o	4.15	4.72	4.33	4.17
o4-mini	GPT-4o	GPT-4o	4.16	4.74	4.34	4.17
GPT-5 mini	GPT-4o	GPT-4o	4.17	4.76	4.37	4.18

mini), the evaluation scores remain highly comparable, with only marginal differences observed. This consistency indicates that the feedback-revision pipeline is robust to backbone variations, ensuring stable performance even when lighter or newer models are employed. These findings highlight the generalisation of the proposed multi-agent architecture and its potential to provide reliable evaluation outcomes across a wide range of foundation models.

Table 5 compares end-to-end agent pipelines constructed using a diverse set of LLM backbones, including GPT-4o, GPT-5 mini, Claude Haiku 4.5,¹² Gemini 2.5 Flash,¹³ and Llama 4 Maverick.¹⁴ In each configuration, RiskAgent, FeedbackAgent, and RevisionAgent adopt the same underlying model, enabling a controlled evaluation of how different LLM backbones influence the overall performance of the MARS framework. This design allows us to assess the behaviour of a broad range of contemporary LLMs and understand how backbone characteristics affect evaluation quality within a unified pipeline.

Results in Table 5 show that the evaluation outcomes produced by MARS remain remarkably consistent across all tested LLM backbones. Although these models differ substantially in design and origin, the scores across

severity accuracy, narrative fidelity, reason clarity, and overall confidence vary only within a narrow range. This stability indicates that the MARS pipeline functions reliably regardless of which underlying model is used. The small performance gap across configurations demonstrates that MARS does not depend on a specific LLM backbone and can maintain coherent and high-quality evaluations even under heterogeneous model choices. These findings confirm the model-agnostic nature of MARS and its robustness across diverse LLM architectures.

Table 6 presents the results of configurations designed to test whether the feedback-revision process depends on using the same underlying LLM for both the FeedbackAgent and RevisionAgent. In this experiment, we introduce controlled substitutions by replacing either the FeedbackAgent or the RevisionAgent with an alternative backbone such as Claude Haiku 4.5, while keeping the remaining agents unchanged. This setup breaks the fully homogeneous configuration and allows us to examine whether the MARS pipeline maintains stable evaluation behaviour when backbone consistency across agents is modified.

The results in Table 6 show that the evaluation outcomes remain highly consistent across all cross-agent

Table 5. Performance comparison of end-to-end agent pipelines across diverse LLM backbones.

Risk	Feedback	Revision	Severity accuracy	Narrative fidelity	Reason clarity	Overall confidence
GPT-4o	GPT-4o	GPT-4o	4.18	4.74	4.26	4.18
GPT-5 mini	GPT-5 mini	GPT-5 mini	4.58	4.85	4.98	4.53
Claude Haiku 4.5	Claude Haiku 4.5	Claude Haiku 4.5	4.02	4.32	4.81	4.01
Gemini 2.5 Flash	Gemini 2.5 Flash	Gemini 2.5 Flash	4.11	4.81	4.72	4.53
Llama 4 Maverick	Llama 4 Maverick	Llama 4 Maverick	4.53	4.99	4.91	4.54

Table 6. Performance comparison of severity accuracy, narrative fidelity, reason clarity, and overall confidence under cross-agent LLM backbone variations.

Risk	Feedback	Revision	Severity accuracy	Narrative fidelity	Reason clarity	Overall confidence
GPT-4o	GPT-4o	GPT-4o	4.18	4.74	4.26	4.18
Claude Haiku 4.5	GPT-4o	GPT-4o	4.25	4.77	4.45	4.27
GPT-4o	Claude Haiku 4.5	GPT-4o	3.90	4.17	4.35	3.86
GPT-4o	GPT-4o	Claude Haiku 4.5	4.27	4.85	4.51	4.28

backbone variations. Replacing the `FeedbackAgent` or `RevisionAgent` with an alternative LLM produces only minor fluctuations in severity accuracy, narrative fidelity, reason clarity, and overall confidence. This stability indicates that the feedback-revision process does not rely on shared backbone identity and that MARS maintains coherent and reliable evaluation behaviour even when the consistency of the underlying models is altered. These findings further support the robustness, flexibility, and model-independence of the MARS pipeline.

6.2.2. Computational efficiency

Table 7 reports the computational efficiency of various `RiskAgent` configurations, measured using three indicators: input tokens, output tokens, and end-to-end run time. Here, input tokens represent the total number of tokens consumed by the prompts provided to the `RiskAgent`, while output tokens correspond to the tokens generated during inference. These quantities together characterise the processing load and generative behaviour of each backbone.

The results show clear differences across LLMs. Models such as GPT-5 mini generate substantially more output tokens due to reasoning tokens, leading to longer run time, whereas backbones like GPT-4o and Gemini 2.5 Flash operate more efficiently with lower token usage and faster execution. Importantly, despite these variations in computational cost, the performance results in Table 5

demonstrate that evaluation quality remains stable across backbones. This indicates that MARS can accommodate LLMs with diverse efficiency profiles, offering flexibility in balancing evaluation quality and computational budget.

6.2.3. Sensitivity analysis

We conduct a sensitivity analysis to examine how the evaluation metrics respond when the severity scale used during assessment is modified. In this analysis, the original 1–5 severity range is replaced with alternative ranges of 1–3 and 1–7, and the corresponding scoring instructions are adjusted accordingly. Table 8 presents the average values for severity accuracy, narrative fidelity, reason clarity, and overall confidence for the three risk levels considered.

The results show that the four metrics remain relatively stable across risk levels 3, 5, and 7, with only modest variations observed among the different configurations. All dimensions exhibit a consistent pattern without abrupt shifts or irregular fluctuations, indicating that adjusting the severity scale and its associated instructions does not introduce instability into the evaluation process. These observations suggest that the multi-agent mechanism is robust to changes in the underlying scoring scheme and maintains predictable behaviour across alternative sensitivity configurations.

Table 7. Performance comparison of input tokens, output tokens, and run time of various `RiskAgent` model configurations.

Risk	Feedback	Revision	Input tokens	Output tokens	Run time (h:m:s)
GPT-4o	GPT-4o	GPT-4o	214,560	28,710	0:22:55
GPT-5 mini	GPT-5 mini	GPT-5 mini	214,302	90,619	0:50:14
Claude Haiku 4.5	Claude Haiku 4.5	Claude Haiku 4.5	241,090	37,932	0:24:46
Gemini 2.5 Flash	Gemini 2.5 Flash	Gemini 2.5 Flash	222,761	29,589	0:25:46
Llama 4 Maverick	Llama 4 Maverick	Llama 4 Maverick	215,681	23,290	0:28:01

Table 8. Sensitivity analysis results across different risk levels for severity accuracy, narrative fidelity, reason clarity, and overall confidence.

Risk levels	Severity accuracy	Narrative fidelity	Reason clarity	Overall confidence
1–3	4.37	4.85	4.53	4.37
1–5	4.18	4.74	4.26	4.18
1–7	4.10	4.69	4.37	4.13

6.2.4. Convergence behaviour

We extend the feedback-revision loop by introducing multiple rounds of iterative refinement and evaluating how the severity assessments and their associated quality metrics evolve across revisions. Specifically, we compute the severity level Mean Absolute Error (MAE) between the severity predicted at each iteration and that from the previous iteration, allowing us to quantify how much each revision step alters the severity output. Table 9 summarises these iterative outcomes, reporting both mean values and standard deviations across all evaluated weather events.

The results in Table 9 show that the feedback-revision loop converges quickly. The severity MAE becomes small after the first revision, indicating that most adjustments occur early in the process. Subsequent iterations introduce only marginal changes, and the MAE steadily approaches zero. The other evaluation metrics, including severity accuracy, narrative fidelity, reason clarity, and overall confidence, exhibit similarly modest variance and stable behaviour across iterations, demonstrating that the pipeline reaches a steady state rapidly. These outcomes suggest that a single revision step is generally sufficient, as additional iterations yield diminishing returns.

6.2.5. Reproducibility

Table 10 evaluates the reproducibility of the MARS framework by executing the full MARS pipeline multiple times under identical settings. Each run follows the same input, configuration, and model backbone, allowing us to isolate natural stochasticity arising from LLM sampling. To quantify run-to-run variation in the assigned severity levels, we report the severity MAE, defined as the mean absolute difference in severity scores between a given run and the reference run. By comparing the evaluation metrics and the severity MAE across repeated executions, we assess whether the MARS framework yields consistent outcomes even when nondeterministic model behaviours are present.

The results demonstrate that MARS exhibits strong reproducibility: severity accuracy, narrative fidelity, reason clarity, and overall confidence remain highly consistent across runs, and the severity MAE values are

small, indicating limited deviation in severity assignments across executions. All observed fluctuations fall within expected LLM variance. These findings show that the MARS framework is stable and reliable, producing nearly identical evaluation outputs across independent runs. This run-to-run consistency further validates the robustness of the proposed framework in practical deployment scenarios.

6.3. Illustrative example of candidate scoring

We present an example centred on a specific target hub at coordinates (33°34'23"N, 86°49'13"W) to illustrate our candidate logistic hub ranking process. Within the vicinity, we evaluate 12 candidate logistic hub locations across a diverse set of dimensions, including spatial distance, annual cost estimates, crime severity index (CSI), and aggregated risks derived from severe weather, traffic accidents, and traffic congestion events.

Table 11 presents the raw, unnormalised values for all 15 scoring dimensions used in the candidate evaluation. These include spatial distance, annual cost, CSI, and severity probabilities for three other risk types: severe weather (5 levels), traffic accidents (4 levels), and traffic congestion (3 levels). These variables differ significantly in scale and units. Distance is measured in miles, cost is expressed in U.S. dollars, CSI is unitless, and three other risk indicators are represented as raw probabilities associated with specific severity levels. For example, 'Weather S4' indicates the yearly probability of a level-4 severe weather event at a candidate location. This detailed hazard structure is essential for capturing nuanced risk exposure but prevents direct aggregation across indicators.

Table 12 presents the normalised risk indicators and corresponding composite scores for all candidate logistic hubs associated with the target location. Each candidate is characterised by six normalised dimensions: distance to the target hub (d), estimated yearly cost (c), CSI (r), and aggregated risk scores for severe weather (s), traffic accidents (a), and traffic congestion (t). These indicators, previously normalised to the $[0, 1]$ range within the candidate set, are combined using root-mean-square (RMS) formulation described in Appendix 1 Equation (A2). This aggregation method ensures that all dimensions contribute proportionally to the final composite score while penalising high risk values. The resulting scores enable transparent, risk-informed ranking of candidate hubs within the target region.

Figure 9 presents the multi-dimensional risk profiles of the top five candidate logistic hubs near the target location (33°34'23"N, 86°49'13"W). Among these, three candidates stand out for their strong performance across

Table 9. Convergence behaviour of evaluation metrics across revision iterations, with mean and standard deviation reported for all metrics.

Revision iteration	Severity MAE	Severity accuracy	Narrative fidelity	Reason clarity	Overall confidence
0	–	4.05 ± 0.42	4.63 ± 0.51	4.23 ± 0.51	4.06 ± 0.41
1	0.125	4.18 ± 0.45	4.68 ± 0.47	4.45 ± 0.53	4.19 ± 0.45
2	0.090	4.19 ± 0.45	4.74 ± 0.46	4.34 ± 0.50	4.22 ± 0.43
3	0.069	4.17 ± 0.43	4.76 ± 0.45	4.35 ± 0.48	4.19 ± 0.41

Table 10. Performance comparison across independent runs of the MARS framework.

Run number	Severity MAE	Severity accuracy	Narrative fidelity	Reason clarity	Overall confidence
#0	–	4.18	4.68	4.45	4.19
#1	0.40	4.15	4.69	4.38	4.16
#2	0.32	4.19	4.72	4.34	4.22
#3	0.36	4.19	4.72	4.38	4.21

key dimensions. The top-ranked hub benefits from its close proximity to the target, along with minimal exposure to severe weather and traffic congestion risks, despite having relatively high crime and traffic accident scores, its overall low risk spatial and environmental profile drives the best composite score. The second-ranked candidate exhibits low crime severity and traffic congestion risk, and although it is farther away and incurs higher cost, its strong safety indicators compensate for these disadvantages. The third-ranked hub shares similar spatial and cost advantages with the top-ranked hub, but its higher severe weather and accident risks result in a slightly lower score. In contrast, lower-ranked candidates such as the 11th- and 12th-ranked hubs show elevated risks across multiple dimensions, particularly distance, congestion, and weather, which outweigh their strengths in cost or crime. These comparisons highlight how top-performing hubs achieve balanced profiles across all dimensions rather than excelling in just one or two.

This example illustrates how the proposed MARS framework enables a holistic assessment of candidate hubs by integrating heterogeneous risk information into a unified scoring process. The comparison of results further reveals clear differences in risk profiles among the candidates. Top-ranked hubs exhibit a balanced profile across multiple dimensions; for example, they are located closer to the target hub, maintain moderate cost, and show low exposure to weather and traffic risks. In contrast, lower-ranked hubs may excel in certain individual aspects such as cost or crime rate, yet suffer from elevated risks in other dimensions such as distance, weather, or congestion, resulting in higher composite scores. These results highlight that the MARS framework provides transparent and interpretable, risk-aware decision support by systematically comparing heterogeneous risk factors across candidate hubs.

To quantitatively evaluate how different risk dimensions contribute to the final logistic hub selection, as

illustrated in Figure 10, we employ an inversion-based ranking consistency analysis. For each target hub, all candidate sites are first ranked according to their final composite scores to establish a baseline order. The same set of candidates is then re-ranked based on individual normalised risk indicators. The number of inversions between each dimension's ranking and the baseline ranking is calculated to measure their degree of disagreement. A smaller inversion count indicates a higher level of consistency with the final ranking, implying that the corresponding factor has a stronger influence on the overall decision. To ensure comparability across dimensions, inversion counts are normalised by their theoretical maximum values, and the normalised scores are averaged across all target hubs. This method reveals the relative impact of each risk dimension in the integrated decision model and provides interpretability for how the final ranking emerges from multi-dimensional risk trade-offs.

The results of this analysis demonstrate clear differences in ranking consistency across the evaluated dimensions. Factors related to traffic congestion and spatial distance show the highest alignment with the final composite ranking, indicating that they exert the strongest influence on the overall decision. Cost and crime severity exhibit moderate consistency, suggesting that they contribute meaningfully but not decisively to the integrated evaluation. In contrast, traffic accident and weather-related risk dimensions show relatively lower consistency, implying that their effects on final site selection are more localised or secondary. Overall, these findings highlight that spatial and operational factors play a more dominant role in shaping the comprehensive ranking outcomes within the MARS framework.

6.4. Human evaluation

To complement the automatic evaluation conducted by the FeedbackAgent and RevisionAgent, a

Table 11. Raw indicators for 12 candidate logistic hubs near the target hub (33°34'23"N, 86°49'13"W), including spatial distance (miles), estimated annual cost (USD), CSI, and severity-specific annual probabilities for severe weather, traffic accidents, and traffic congestion.

Candidate	Dist (mi)	Cost (\$)	CSI	Weather S1	Weather S2
(33°30'22"N, 86°48'50"W)	7.47	190,000	2641.27	0.963	1.000
(33°30'08"N, 86°49'26"W)	7.90	162,037	2641.27	0.963	1.000
(33°39'13"N, 86°42'48"W)	13.36	107,187	2641.27	0.977	1.000
(33°19'52"N, 86°54'45"W)	28.24	91,719	3661.40	0.958	1.000
(33°17'07"N, 86°48'41"W)	32.02	107,968	254.42	0.957	1.000
(33°14'31"N, 87°35'16"W)	80.19	232,750	678.46	0.923	1.000
(34°18'22"N, 86°35'51"W)	84.05	526,050	497.00	0.953	1.000
(34°36'20"N, 86°45'25"W)	114.96	1,297,620	177.04	0.949	1.000
(34°38'36"N, 86°54'37"W)	119.29	989,827	351.66	0.947	1.000
(34°42'08"N, 86°41'30"W)	126.09	125,361	204.71	0.929	1.000
(34°42'48"N, 86°36'31"W)	128.27	78,450	204.71	0.926	1.000
(34°46'54"N, 86°39'45"W)	135.17	62,0000	204.71	0.919	1.000
Candidate	Weather S3	Weather S4	Weather S5	Acc S1	Acc S2
(33°30'22"N, 86°48'50"W)	1.000	0.998	0.000	0.000	0.000
(33°30'08"N, 86°49'26"W)	1.000	0.998	0.000	0.000	0.000
(33°39'13"N, 86°42'48"W)	1.000	0.999	1.000	1.000	1.000
(33°19'52"N, 86°54'45"W)	1.000	0.999	0.000	0.000	0.000
(33°17'07"N, 86°48'41"W)	1.000	0.999	1.000	1.000	1.000
(33°14'31"N, 87°35'16"W)	1.000	0.998	1.000	1.000	0.938
(34°18'22"N, 86°35'51"W)	1.000	0.997	0.000	0.000	0.000
(34°36'20"N, 86°45'25"W)	1.000	0.988	0.000	0.000	0.000
(34°38'36"N, 86°54'37"W)	1.000	0.986	0.000	0.000	0.000
(34°42'08"N, 86°41'30"W)	1.000	0.991	1.000	1.000	1.000
(34°42'48"N, 86°36'31"W)	1.000	0.991	0.000	0.000	0.000
(34°46'54"N, 86°39'45"W)	1.000	0.992	1.000	1.000	1.000
Candidate	Acc S3	Acc S4	Congestion S1	Congestion S2	Congestion S3
(33°30'22"N, 86°48'50"W)	0.000	0.000	0.000	0.000	0.000
(33°30'08"N, 86°49'26"W)	0.000	1.000	1.000	0.750	0.000
(33°39'13"N, 86°42'48"W)	0.000	1.000	1.000	0.750	0.000
(33°19'52"N, 86°54'45"W)	0.000	1.000	1.000	0.750	0.000
(33°17'07"N, 86°48'41"W)	1.000	1.000	1.000	1.000	0.000
(33°14'31"N, 87°35'16"W)	1.000	1.000	1.000	1.000	0.000
(34°18'22"N, 86°35'51"W)	0.000	1.000	1.000	0.500	0.000
(34°36'20"N, 86°45'25"W)	1.000	1.000	1.000	0.788	0.500
(34°38'36"N, 86°54'37"W)	0.253	1.000	1.000	0.783	0.000
(34°42'08"N, 86°41'30"W)	1.000	1.000	1.000	0.788	0.500
(34°42'48"N, 86°36'31"W)	1.000	1.000	1.000	0.788	0.500
(34°46'54"N, 86°39'45"W)	1.000	1.000	1.000	0.788	0.500

human evaluation protocol is designed to establish an external benchmark for assessing the reliability and interpretability of the proposed multi-agent framework. A total of 200 severe weather event narratives are randomly sampled from the NOAA Storm Events database. Each narrative is independently evaluated by at least two authors with relevant expertise in meteorology, risk analysis, or supply chain management.

Annotators rate each narrative along three dimensions: *Severity* (1–5), *Clarity* (1–3), and *Confidence* (1–3), by following instructions in Appendix 3. The severity score represents the overall impact level described in the text, the clarity score measures how well the narrative conveys the impact, and the confidence score indicates the annotator's certainty. On average, the clarity and confidence ratings are 2.11 (standard deviation $\sigma = 0.52$) and 2.21 ($\sigma = 0.41$), respectively, indicating that most narratives are reasonably detailed and the annotators are

confident in their assessments. All judgments are made solely based on the given narratives without consulting external data.

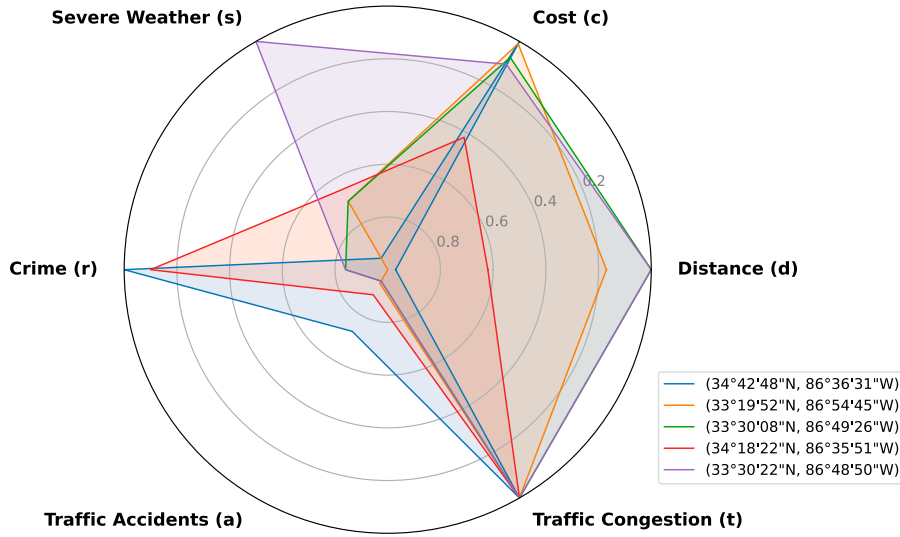
Aggregated human ratings are used as a benchmark to compare with model-generated severity scores. The rule-based baseline (Appendix 4) derives event severity from keyword frequency and predefined linguistic patterns in NOAA event narratives, representing a deterministic scoring approach without contextual reasoning. Figure 11 compares the human-assigned average severity with generation from this rule-based baseline and from the LLM after revision. Each dot represents a unique event narrative, and dot size reflects the frequency of overlapping ratings. The diagonal line ($y = x$) indicates perfect agreement between human and model scores.

The results demonstrate a stronger alignment between human ratings and LLM generation after revision, indicating that the feedback-revision loop improves both

Table 12. Normalised risk indicators, composite scores, and ranks across six dimensions for candidate logistic hubs targeting the hub at (33°34'23"N, 86°49'13"W) (lower number is better, and larger number is worse).

Candidate	Rank	Score	Distance	Cost	Crime	Weather	Accidents	Congestion
(33°30'22"N, 86°48'50"W)	1	0.52	0.00	0.10	0.84	0.00	0.95	0.00
(34°18'22"N, 86°35'51"W)	2	0.55	0.62	0.42	0.10	0.65	0.89	0.00
(33°30'08"N, 86°49'26"W)	3	0.59	0.00	0.07	0.84	0.70	0.95	0.00
(33°19'52"N, 86°54'45"W)	4	0.63	0.17	0.01	1.00	0.70	0.94	0.00
(34°42'48"N, 86°36'31"W)	4	0.63	0.97	0.00	0.00	0.95	0.73	0.00
(34°38'36"N, 86°54'37"W)	6	0.66	0.90	0.87	0.05	0.71	0.76	0.00
(33°17'07"N, 86°48'41"W)	7	0.67	0.20	0.02	0.02	0.89	0.94	1.00
(33°14'31"N, 87°35'16"W)	8	0.69	0.59	0.14	0.17	0.89	0.85	0.98
(33°39'13"N, 86°42'48"W)	9	0.73	0.05	0.02	0.84	0.70	1.00	1.00
(34°36'20"N, 86°45'25"W)	9	0.73	0.87	1.00	0.00	0.95	0.77	0.00
(34°42'08"N, 86°41'30"W)	11	0.75	0.96	0.04	0.00	0.95	0.73	1.00
(34°46'54"N, 86°39'45"W)	12	0.79	1.00	0.51	0.00	0.95	0.73	1.00

Note: The six dimensions include distance (*d*), cost (*c*), crime severity (*r*), and aggregated risk scores for severe weather (*s*), traffic accidents (*a*), and traffic congestion (*t*).

**Figure 9.** Multi-dimensional risk profile of top-5 candidate logistic hubs near the target hub (33°34'23"N, 86°49'13"W), based on six normalised indicators: distance, cost, crime, severe weather, traffic accidents, and traffic congestion.

Note: The radar chart scale is inverted (outer = 0, inner = 1), but larger areas still indicate better performance since lower indicator values are preferable.

consistency and agreement with human reasoning compared with the rule-based method.

7. Conclusion

In this paper, we introduce MARS, a novel large language model (LLM) enhanced Multi-Agent framework for Risk assessment in Supply chain, described and experimented in the context of logistic hub site selection. Unlike traditional methods that rely solely on structured metrics, our approach integrates structured indicators with unstructured event narratives. These narratives are semantically processed by LLMs using carefully designed prompts to extract contextual meaning and assign event severities. The transformation is managed by a coordinated multi-agent system: *RiskAgent* performs the initial extraction and scoring, *FeedbackAgent* evaluates the consistency and validity of these outputs, and *RevisionAgent* refines the output through an

iterative feedback-revision loop mechanism. This agent loop ensures that narrative inputs are converted into reliable and interpretable risk indicators. The resulting severity levels are then aggregated with other risk indicators, enabling adaptive, transparent, risk-aware ranking and selection of candidate logistic hubs.

Through extensive assessments across multiple south-eastern U.S. states, including Alabama, Florida, Georgia, Mississippi, North Carolina, South Carolina, and Tennessee, we evaluate hundreds of candidate logistic hub locations. The MARS framework processes thousands of real-world severe weather event narratives, combining this unstructured information with structured indicators such as distance, cost, and crime rates. This large-scale empirical analysis demonstrates the system's capacity to extract and integrate multi-source risk factors for practical use. The ranking mechanism not only identifies the most resilient hub locations, but also offers interpretable insight into why certain sites are preferred, highlighting

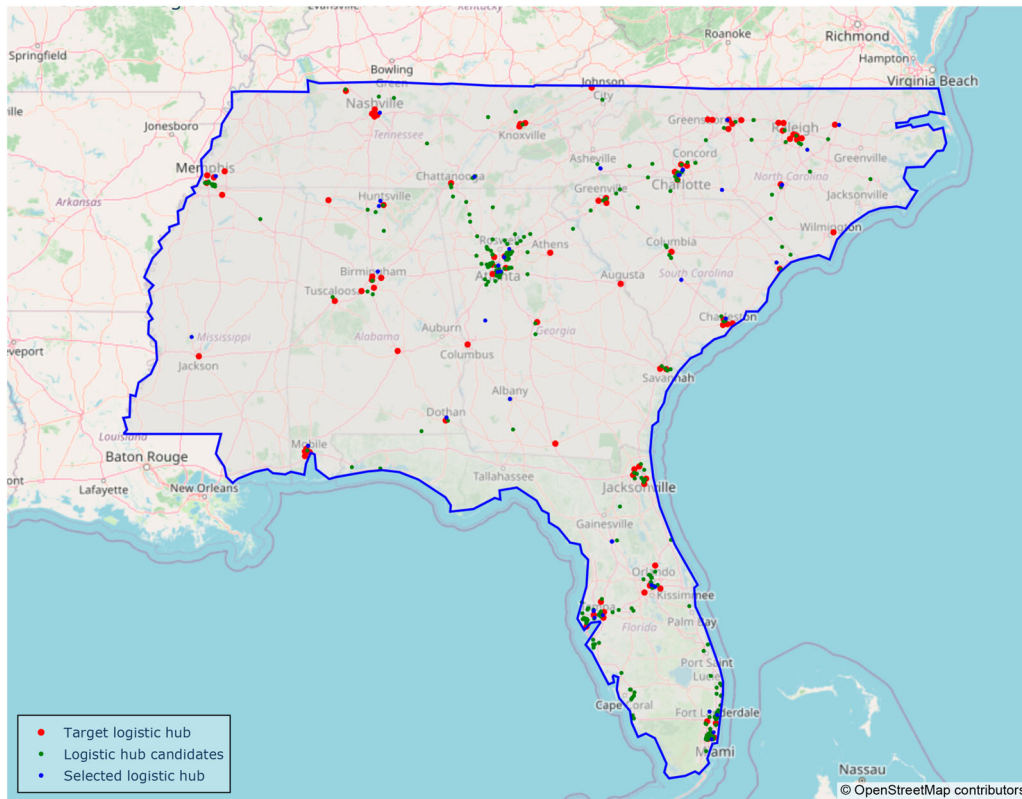


Figure 10. Distribution of selected logistic hubs in the southeastern United States.

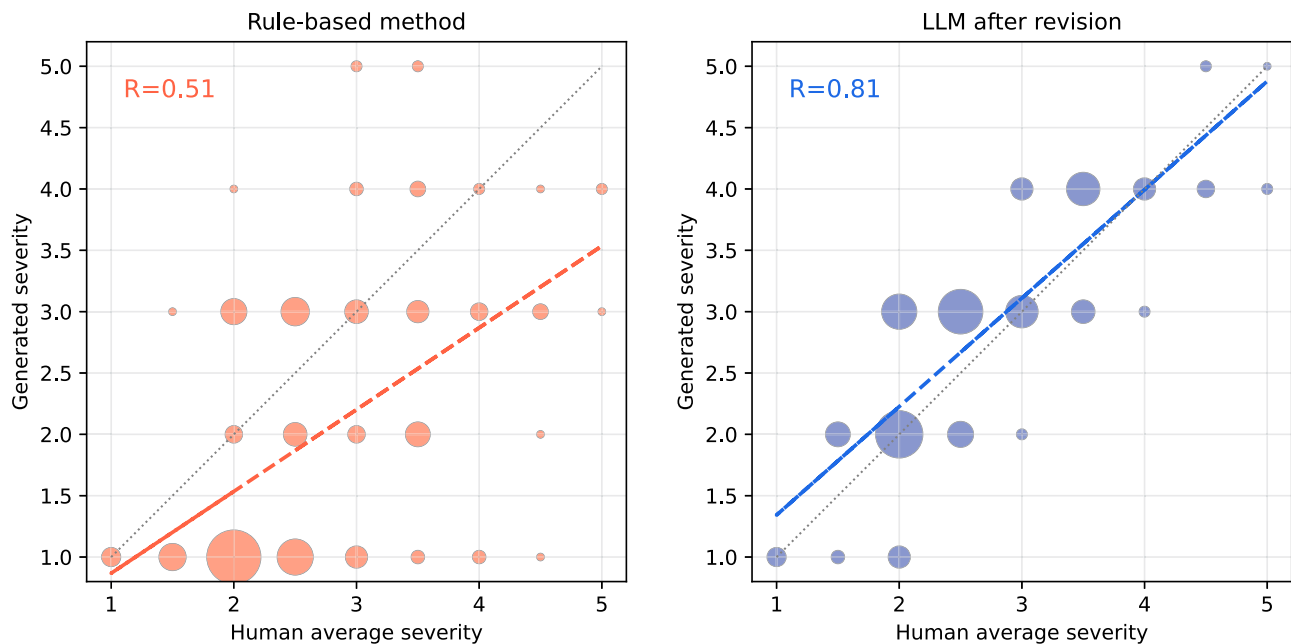


Figure 11. Comparison of human-assigned and model-generated severity. Left: rule-based method ($R = 0.51$). Right: LLM after revision ($R = 0.81$). R denotes the Pearson correlation coefficient computed for each subplot. The diagonal line ($y = x$) indicates perfect agreement, and dot size reflects rating frequency.

trade-offs across six critical risk dimensions, including geographic distance, operational cost, crime severity, and aggregated risks from severe weather, traffic accidents, and traffic congestion among different severities. These

results underscore the MARS framework's utility in supporting data-driven decision-making for resilient supply chain infrastructure. Although this study focuses on the southeastern U.S., MARS itself is broadly generalisable,

as it relies on widely available structured indicators and unstructured event narratives. This flexibility ensures its applicability across diverse geographic and operational contexts without substantial modification.

Our findings demonstrate that incorporating unstructured risk data, such as severe weather narratives, markedly enhances the granularity and contextual relevance of hub site evaluations. For example, some candidate hubs with favorable structured metrics like low operational cost and minimal crime may nonetheless be flagged as high risk due to recurring extreme weather hazards that are only revealed through narrative analysis. This underscores the critical importance of integrating localised, semantically rich assessments of environmental risk into the logistics planning process. While our current work focuses on weather-related narratives, the same approach can be extended to other unstructured risk sources, such as reports of social unrest, community opposition, and other risks expressed in textual form, further broadening the MARS framework's applicability. Additionally, the multi-agent reasoning loop offers a flexible and interpretable mechanism for refining severity scores through iterative feedback and revision, enabling the system to adapt to evolving data and context. These capabilities are vital for real-world deployment, where decision-making must be not only data-driven, but also resilient, transparent, and responsive to domain-specific risk complexities. Overall, the proposed framework lays a foundation for next-generation, LLM-powered supply chain and infrastructure planning tools that can systematically bridge structured analytics with unstructured intelligence.

Notes

1. <https://www.costar.com/>
2. <https://www.ncdc.noaa.gov/stormevents/>
3. <https://www.ncei.noaa.gov/>
4. <https://github.com/langchain-ai/langgraph>
5. <https://ops.fhwa.dot.gov/publications/fhwahop20028/index.htm>
6. <https://cde.ucr.cjis.gov/LATEST/webapp/#/pages/downloads>
7. <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.BallTree.html>
8. https://www.statcan.gc.ca/en/statistical-programs/document/3302_D16_V1
9. <https://platform.openai.com/docs/models/gpt-4o>
10. <https://platform.openai.com/docs/models/o4-mini>
11. <https://platform.openai.com/docs/models/gpt-5-mini>
12. <https://openrouter.ai/anthropic/claude-haiku-4.5>
13. <https://openrouter.ai/google/gemini-2.5-flash>
14. <https://openrouter.ai/meta-llama/llama-4-maverick>

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are available upon request from the corresponding author, Yinzhu Quan.

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Appendices

Appendix 1. Risk aggregation method

We begin by applying quantile normalisation to each risk dimension separately within the set of candidates associated with a given target hub. For each dimension, values below the 5th percentile are assigned a normalised score of 0, those above the 95th percentile are assigned a score of 1, and intermediate values are scaled linearly. This transformation ensures comparability across heterogeneous indicators.

To aggregate the normalised risk scores, we first compute a sub-score for each individual hazard category, namely severe weather, traffic accidents, and traffic congestion. Each category consists of multiple severity levels, and each candidate hub has an associated normalised probability for each level. To summarise these into a single hazard score per category, we compute a weighted root-mean-square (RMS) of the severity probabilities. The weights are chosen to reflect the relative importance of higher severity levels, greater severity levels are assigned larger weights. For a given hazard type h with severity levels $s_1, s_2, \dots, s_n$ and corresponding normalised probabilities $p_1, p_2, \dots, p_n$, the aggregated hazard score is computed as:

$$\text{HazardScore}_h = \frac{1}{\sqrt{\sum_{i=1}^n w_i}} \cdot \sqrt{\sum_{i=1}^n w_i \cdot p_i^2} \quad (\text{A1})$$

where w_i is the weight assigned to severity level s_i .

Once individual scores for severe weather, traffic accidents, and traffic congestion are computed, we combine them with the normalised cost, distance, and crime severity values to form a unified composite score. This final score also uses a RMS formulation:

$$\text{Composite Score} = \frac{1}{\sqrt{6}} \cdot \sqrt{d^2 + c^2 + r^2 + s^2 + a^2 + t^2} \quad (\text{A2})$$

Here, we assign equal weights to all six risk dimensions in the RMS formulation, assuming equal importance in the absence of domain-specific preferences or empirical evidence. Each variable corresponds to a normalised score or risk component:

- d : normalised distance to the target hub
- c : normalised estimated annual cost
- r : normalised crime severity index
- s : aggregated severe weather risk score
- a : aggregated traffic accident risk score
- t : aggregated traffic congestion risk score

All features are normalised to the range $[0, 1]$ across each target hub's candidate set. The final score reflects a balanced combination of infrastructure cost, spatial distance, and multi-hazard risk. This composite scoring scheme enables robust and consistent comparison of candidate hub locations across all target regions.

Appendix 2. Examples of different severity levels

Examples of event narratives, their assigned severity levels, and corresponding reasoning can be seen in Table A1.

Table A1. Examples of event descriptions by severity level.

Event type	Event narrative	Severity level	Reason
Thunderstorm Wind	Emergency manager reported a tree down blocking one lane on Tails Creek Rd near the intersection with Pleasant Grove Rd.	1	A single tree down blocking one lane does not cause major disruptions, justifying a level 1 severity.
Flood	Approximately 4-6 inches of water over roadways to the West of downtown Athens. Most likely nuisance flooding as heavy rainfall moved through the area.	2	Minor flooding occurred with no significant infrastructure damage, appropriate for a severity level of 2.
Hail	About 2 inch hail and golf ball sized hail were observed near Camp Blanding.	3	Hail size up to 2 inches and golf ball-sized hail can cause moderate impact, including possible damage to vehicles and structures, which supports a severity level of 3.
Flash Flood	City of Conway reports multiple road closures around the city due to flash flooding, including portions of Highways 701, 501, and 378.	4	High impact due to widespread road closures affecting major highways, significantly disrupting transportation and accessibility.
Heavy Rain	Mesonet site recorded 15.23 inches of rain in the last 6 hours.	5	The extremely high rainfall of 15.23 inches in just 6 hours indicates a critical risk of severe flooding and significant transport disruption, warranting the highest severity level.

Appendix 3. Human evaluation instructions

This appendix provides the instructions given to human evaluators for assigning severity ratings to event narratives.

Human Evaluation Instructions

You are an assistant specialized in evaluating weather-related risks for logistic hubs. Given the following event narratives extracted from the NOAA Storm Events database, your task is to assess the impact severity of each weather event based solely on the textual description. You should not infer additional information beyond what is explicitly stated in the text.

For each event narrative, assign three ratings:

1. Severity (1-5): overall seriousness or impact level of the event.

1: Minimal impact, no significant disruptions, minor weather effects such as light rain or wind.

2: Low impact, localized disturbances, minor flooding, or moderate wind speeds causing slight delays.

3: Moderate impact, disruptions to operations, road closures, structural damage, or power outages in some areas.

4: High impact, significant disruptions, widespread infrastructure damage, major flooding, or high wind speeds causing logistical challenges.

5: Severe impact, extreme weather conditions such as hurricanes or tornadoes, widespread destruction, long-term operational shutdowns.

2. Clarity (1-3): how clearly the event's impact is described in the narrative.

1: Unclear. The description lacks essential details or is ambiguous.

2: Moderately clear. Provides some impact information but lacks specificity.

3: Clear. The impact description is detailed, concrete, and easy to interpret.

3. Confidence (1-3): how certain you feel about your own severity rating.

1: Low confidence. The text is too vague to decide confidently.

2: Medium confidence. Some uncertainty remains.

3: High confidence. The description is clear and the rating feels certain.

Read each narrative carefully and assign one score for each dimension. Use integer values only and rate each narrative independently. Base your judgments entirely on the given text without referencing external data. Your evaluations will serve as human ground truth for validating the LLM-based multi-agent risk assessment framework.

Appendix 4. Rule-based severity scoring method

The rule-based severity scoring method is developed to provide an interpretable baseline for assessing weather event impact levels based solely on textual descriptions. This approach relies on a curated lexicon of representative keywords and phrases associated with increasing degrees of weather-related disruption. Each narrative is scanned for the presence of these terms, which are grouped into five severity categories ranging from minimal to severe impact. The category with the highest frequency of matching terms determines the assigned score; in the case of ties, the median severity level is selected. The complete set of keywords used for rule-based evaluation is summarised in Table A2, which outlines the linguistic indicators corresponding to each severity level.

Table A2. Representative keywords and phrases for each severity level.

Severity level	Representative keywords/phrases
1 (Minimal Impact)	hail, drizzle, light rain, cloud, funnel, measured, observer, public report, minor, visibility, gust, isolated showers, trace precipitation, sprinkles, brief rain
2 (Low Impact)	localised flooding, pooling, minor flooding, small hail, trees down, gusty winds, thunderstorm, brief storm, street flooding, standing water, minor damage, rainfall, isolated flooding, brief outage, blown debris
3 (Moderate Impact)	tornado, damage, uprooted, snapped, power line, outage, roof damage, flooding, blocked road, debris, strong winds, downed trees, structural damage, heavy rain, severe storm
4 (High Impact)	widespread, major flooding, homes damaged, flash flooding, road closures, vehicles stranded, emergency response, multiple homes, severe thunderstorm, significant damage, collapsed, infrastructure, extensive trees down, evacuation, flash flood
5 (Severe Impact)	ef, destroyed, fatalities, catastrophic, widespread destruction, neighbourhoods levelled, long-term outage, mobile homes, collapse, infrastructure failure, disaster, emergency declared, devastation, extreme winds, severe tornado